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Augmented-reality surgical system using depth sensing

Abstract

Disclosed herein are systems, devices, and methods for image-guided surgery. Some systems include a head-mounted unit, having a see-through augmented-reality display and a depth sensor, which is configured to generate depth data with respect to a region of interest (ROI) of a body of a patient that is viewed through the display by a user wearing the head-mounted unit. A processor, which is configured to receive a three-dimensional (3D) tomographic image of the body of the patient, computes a depth map of the ROI based on the depth data generated by the depth sensor, to compute a transformation over the ROI so as to register the tomographic image with the depth map, and to apply the transformation in presenting a part of the tomographic image on the display in registration with the ROI viewed through the display.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3101715	12/1962	Glassman	N/A	N/A
3690776	12/1971	Zaporoshan	N/A	N/A
4459358	12/1983	Berke	N/A	N/A
4711512	12/1986	Upatnieks	N/A	N/A
4863238	12/1988	Brewster	N/A	N/A
4944739	12/1989	Torre	N/A	N/A
5100420	12/1991	Green et al.	N/A	N/A
5147365	12/1991	Whitlock et al.	N/A	N/A
5357292	12/1993	Wiedner	N/A	N/A
5410802	12/1994	Buckley	N/A	N/A
5441042	12/1994	Putman	N/A	N/A
5442146	12/1994	Bell et al.	N/A	N/A
5510832	12/1995	Garcia	N/A	N/A

D370309	12/1995	Stucky	N/A	N/A
5620188	12/1996	McCurry et al.	N/A	N/A
5636255	12/1996	Ellis	N/A	N/A
5665092	12/1996	Mangiardi et al.	N/A	N/A
5743731	12/1997	Lares et al.	N/A	N/A
5771121	12/1997	Hentschke	N/A	N/A
5792046	12/1997	Dobrovolny	N/A	N/A
5841507	12/1997	Barnes	N/A	N/A
6006126	12/1998	Cosman	N/A	N/A
6038467	12/1999	De Bliek et al.	N/A	N/A
6125164	12/1999	Murphy et al.	N/A	N/A
6138530	12/1999	McClure	N/A	N/A
6147805	12/1999	Ferguson	N/A	N/A
6227667	12/2000	Halldorsson et al.	N/A	N/A
6256529	12/2000	Holupka et al.	N/A	N/A
6285505	12/2000	Melville et al.	N/A	N/A
6314310	12/2000	Ben-Haim et al.	N/A	N/A
6349001	12/2001	Spitzer	N/A	N/A
6444192	12/2001	Mattrey	N/A	N/A
6447503	12/2001	Wynne et al.	N/A	N/A
6449090	12/2001	Omar et al.	N/A	N/A
6456405	12/2001	Horikoshi et al.	N/A	N/A
6456868	12/2001	Saito et al.	N/A	N/A
6474159	12/2001	Foxlin et al.	N/A	N/A
6518939	12/2002	Kikuchi	N/A	N/A
6527777	12/2002	Justin	N/A	N/A
6529331	12/2002	Massof et al.	N/A	N/A
6549645	12/2002	Oikawa et al.	N/A	N/A
6578962	12/2002	Amir et al.	N/A	N/A
6609022	12/2002	Vilsmeier et al.	N/A	N/A
6610009	12/2002	Person	N/A	N/A
D480476	12/2002	Martinson et al.	N/A	N/A
6659611	12/2002	Amir et al.	N/A	N/A
6675040	12/2003	Cosman	N/A	N/A
6683584	12/2003	Ronzani et al.	N/A	N/A
6690964	12/2003	Bieger et al.	N/A	N/A
6714810	12/2003	Grzeszczuk et al.	N/A	N/A
6737425	12/2003	Yamamoto et al.	N/A	N/A
6740882	12/2003	Weinberg	N/A	N/A
6757068	12/2003	Foxlin	N/A	N/A
6759200	12/2003	Stanton, Jr.	N/A	N/A
6847336	12/2004	Lemelson et al.	N/A	N/A
6856324	12/2004	Sauer et al.	N/A	N/A
6856826	12/2004	Seeley et al.	N/A	N/A
6891518	12/2004	Sauer et al.	N/A	N/A
6900777	12/2004	Hebert et al.	N/A	N/A
6919867	12/2004	Sauer	N/A	N/A
6921167	12/2004	Nagata	N/A	N/A
6966668	12/2004	Cugini et al.	N/A	N/A
6980849	12/2004	Sasso	N/A	N/A

6993374	12/2005	Sasso	N/A	N/A
6997552	12/2005	Hung	N/A	N/A
6999239	12/2005	Martins et al.	N/A	N/A
7000262	12/2005	Bielefeld	N/A	N/A
7035371	12/2005	Boese et al.	N/A	N/A
7043961	12/2005	Pandey et al.	N/A	N/A
7072435	12/2005	Metz et al.	N/A	N/A
7103233	12/2005	Stearns	N/A	N/A
7107091	12/2005	Jutras et al.	N/A	N/A
7112656	12/2005	Desnoyers et al.	N/A	N/A
7141812	12/2005	Appleby et al.	N/A	N/A
7157459	12/2006	Ohta et al.	N/A	N/A
7169785	12/2006	Timmer et al.	N/A	N/A
7171255	12/2006	Holupka et al.	N/A	N/A
7176936	12/2006	Sauer et al.	N/A	N/A
7187792	12/2006	Fu et al.	N/A	N/A
7190331	12/2006	Genc et al.	N/A	N/A
7194295	12/2006	Vilsmeier	N/A	N/A
7215322	12/2006	Genc et al.	N/A	N/A
7229078	12/2006	Lechot	N/A	N/A
7231076	12/2006	Fu et al.	N/A	N/A
7235076	12/2006	Pacheco	N/A	N/A
7239330	12/2006	Sauer et al.	N/A	N/A
7241292	12/2006	Hooven	N/A	N/A
7259266	12/2006	Carter et al.	N/A	N/A
7260426	12/2006	Schweikard et al.	N/A	N/A
7269192	12/2006	Hayashi	N/A	N/A
7281826	12/2006	Huang	N/A	N/A
7315636	12/2007	Kuduvalli	N/A	N/A
7320556	12/2007	Vagn-Erik	N/A	N/A
7330578	12/2007	Wang et al.	N/A	N/A
7359535	12/2007	Salla et al.	N/A	N/A
7364314	12/2007	Nilsen et al.	N/A	N/A
7366934	12/2007	Narayan et al.	N/A	N/A
7379077	12/2007	Bani-Hashemi et al.	N/A	N/A
7431453	12/2007	Hogan	N/A	N/A
7435219	12/2007	Kim	N/A	N/A
7450743	12/2007	Sundar et al.	N/A	N/A
7458977	12/2007	McGinley et al.	N/A	N/A
7462852	12/2007	Appleby et al.	N/A	N/A
7493153	12/2008	Ahmed et al.	N/A	N/A
7505617	12/2008	Fu et al.	N/A	N/A
7507968	12/2008	Wollenweber et al.	N/A	N/A
7518136	12/2008	Appleby et al.	N/A	N/A
7525735	12/2008	Sottolare et al.	N/A	N/A
D592691	12/2008	Chang	N/A	N/A
D592692	12/2008	Chang	N/A	N/A
D592693	12/2008	Chang	N/A	N/A
7536216	12/2008	Geiger et al.	N/A	N/A
7542791	12/2008	Mire et al.	N/A	N/A

7556428	12/2008	Sukovic et al.	N/A	N/A
7557824	12/2008	Holliman	N/A	N/A
7563228	12/2008	Ma et al.	N/A	N/A
7567834	12/2008	Clayton et al.	N/A	N/A
7570791	12/2008	Frank et al.	N/A	N/A
7586686	12/2008	Hall	N/A	N/A
D602620	12/2008	Cristoforo	N/A	N/A
7605826	12/2008	Sauer	N/A	N/A
7606613	12/2008	Simon et al.	N/A	N/A
7607775	12/2008	Hermanson et al.	N/A	N/A
7620223	12/2008	Xu et al.	N/A	N/A
7623902	12/2008	Pacheco	N/A	N/A
7627085	12/2008	Boyden et al.	N/A	N/A
7630753	12/2008	Simon et al.	N/A	N/A
7633501	12/2008	Wood et al.	N/A	N/A
7645050	12/2009	Wilt et al.	N/A	N/A
7653226	12/2009	Guhring et al.	N/A	N/A
7657075	12/2009	Viswanathan	N/A	N/A
7689019	12/2009	Boese et al.	N/A	N/A
7689042	12/2009	Brunner et al.	N/A	N/A
7689320	12/2009	Prisco et al.	N/A	N/A
7699486	12/2009	Beiner	N/A	N/A
7699793	12/2009	Goette et al.	N/A	N/A
7719769	12/2009	Sugihara et al.	N/A	N/A
D617825	12/2009	Chang	N/A	N/A
7734327	12/2009	Colquhoun	N/A	N/A
D619285	12/2009	Cristoforo	N/A	N/A
7751865	12/2009	Jascob et al.	N/A	N/A
7758204	12/2009	Klipstein et al.	N/A	N/A
7768702	12/2009	Hirose et al.	N/A	N/A
7769236	12/2009	Fiala	N/A	N/A
7773074	12/2009	Arenson et al.	N/A	N/A
7774044	12/2009	Sauer et al.	N/A	N/A
7822483	12/2009	Stone et al.	N/A	N/A
D628307	12/2009	Krause-Bonte	N/A	N/A
7826902	12/2009	Stone et al.	N/A	N/A
7831073	12/2009	Fu et al.	N/A	N/A
7831096	12/2009	Williamson, Jr.	N/A	N/A
7835778	12/2009	Foley et al.	N/A	N/A
7835784	12/2009	Mire et al.	N/A	N/A
7837987	12/2009	Shi et al.	N/A	N/A
7840093	12/2009	Fu et al.	N/A	N/A
7840253	12/2009	Tremblay et al.	N/A	N/A
7840256	12/2009	Lakin et al.	N/A	N/A
7853305	12/2009	Simon et al.	N/A	N/A
7854705	12/2009	Pawluczyk et al.	N/A	N/A
7857271	12/2009	Lees	N/A	N/A
7860282	12/2009	Boese et al.	N/A	N/A
D630766	12/2010	Harbin	N/A	N/A
7865269	12/2010	Prisco et al.	N/A	N/A

7874686	12/2010	Rossner et al.	N/A	N/A
7881770	12/2010	Melkent et al.	N/A	N/A
7893413	12/2010	Appleby et al.	N/A	N/A
7894649	12/2010	Fu et al.	N/A	N/A
7920162	12/2010	Masini et al.	N/A	N/A
7922391	12/2010	Essenreiter et al.	N/A	N/A
7938553	12/2010	Beiner	N/A	N/A
7945310	12/2010	Gattani et al.	N/A	N/A
7953471	12/2010	Clayton et al.	N/A	N/A
7969383	12/2010	Eberl et al.	N/A	N/A
7974677	12/2010	Mire et al.	N/A	N/A
7985756	12/2010	Barlow et al.	N/A	N/A
7991557	12/2010	Liew et al.	N/A	N/A
7993353	12/2010	Roner et al.	N/A	N/A
7996064	12/2010	Simon et al.	N/A	N/A
8004524	12/2010	Deinzer	N/A	N/A
8021300	12/2010	Ma et al.	N/A	N/A
8022984	12/2010	Cheong et al.	N/A	N/A
8045266	12/2010	Nakamura	N/A	N/A
8060181	12/2010	Rodriguez et al.	N/A	N/A
8068581	12/2010	Boese et al.	N/A	N/A
8068896	12/2010	Daghighian et al.	N/A	N/A
8077943	12/2010	Williams et al.	N/A	N/A
8079957	12/2010	Ma et al.	N/A	N/A
8081812	12/2010	Kreiser	N/A	N/A
8085075	12/2010	Huffman et al.	N/A	N/A
8085897	12/2010	Morton	N/A	N/A
8090175	12/2011	Fu et al.	N/A	N/A
8092400	12/2011	Warkentine et al.	N/A	N/A
8108072	12/2011	Zhao et al.	N/A	N/A
8112292	12/2011	Simon	N/A	N/A
8116847	12/2011	Gattani et al.	N/A	N/A
8120847	12/2011	Chang	N/A	N/A
8121255	12/2011	Sugiyama	N/A	N/A
8155479	12/2011	Hoffman et al.	N/A	N/A
8180132	12/2011	Gorges et al.	N/A	N/A
8180429	12/2011	Sasso	N/A	N/A
8208599	12/2011	Ye et al.	N/A	N/A
8216211	12/2011	Mathis et al.	N/A	N/A
8221402	12/2011	Francischelli et al.	N/A	N/A
8239001	12/2011	Verard et al.	N/A	N/A
8244012	12/2011	Liang et al.	N/A	N/A
8253778	12/2011	Atsushi	N/A	N/A
8271069	12/2011	Jascob et al.	N/A	N/A
8280491	12/2011	Kuduvalli et al.	N/A	N/A
8285021	12/2011	Boese et al.	N/A	N/A
8300315	12/2011	Kobayashi	N/A	N/A
8305685	12/2011	Heine et al.	N/A	N/A
8306305	12/2011	Porat et al.	N/A	N/A
8309932	12/2011	Haselman et al.	N/A	N/A

8317320	12/2011	Huang	N/A	N/A
8328815	12/2011	Farr et al.	N/A	N/A
8335553	12/2011	Rubner et al.	N/A	N/A
8335557	12/2011	Maschke	N/A	N/A
8340379	12/2011	Razzaque et al.	N/A	N/A
8369925	12/2012	Giesel et al.	N/A	N/A
8386022	12/2012	Jutras et al.	N/A	N/A
8394144	12/2012	Zehavi et al.	N/A	N/A
8398541	12/2012	Dimaio et al.	N/A	N/A
8444266	12/2012	Waters	N/A	N/A
8457719	12/2012	Moctezuma De La Barrera et al.	N/A	N/A
8467851	12/2012	Mire et al.	N/A	N/A
8469902	12/2012	Dick et al.	N/A	N/A
8475470	12/2012	Von Jako	N/A	N/A
8494612	12/2012	Vetter et al.	N/A	N/A
8509503	12/2012	Nahum et al.	N/A	N/A
8511827	12/2012	Hua et al.	N/A	N/A
8531394	12/2012	Maltz	N/A	N/A
8540364	12/2012	Waters	N/A	N/A
8545012	12/2012	Waters	N/A	N/A
8548567	12/2012	Maschke et al.	N/A	N/A
8556883	12/2012	Saleh	N/A	N/A
8559596	12/2012	Thomson et al.	N/A	N/A
8567945	12/2012	Waters	N/A	N/A
8571353	12/2012	Watanabe	N/A	N/A
8585598	12/2012	Razzaque et al.	N/A	N/A
8600001	12/2012	Schweizer	N/A	N/A
8600477	12/2012	Beyar et al.	N/A	N/A
8605199	12/2012	Imai	N/A	N/A
8611988	12/2012	Miyamoto	N/A	N/A
8612024	12/2012	Stone et al.	N/A	N/A
8634897	12/2013	Simon et al.	N/A	N/A
8641621	12/2013	Razzaque et al.	N/A	N/A
8643950	12/2013	König	N/A	N/A
8644907	12/2013	Hartmann et al.	N/A	N/A
8674902	12/2013	Park et al.	N/A	N/A
8686923	12/2013	Eberl et al.	N/A	N/A
8690581	12/2013	Ruf et al.	N/A	N/A
8690776	12/2013	Razzaque et al.	N/A	N/A
8692845	12/2013	Fedorovskaya et al.	N/A	N/A
8693632	12/2013	Allison	N/A	N/A
8694075	12/2013	Groszmann et al.	N/A	N/A
8699765	12/2013	Hao et al.	N/A	N/A
8705829	12/2013	Frank et al.	N/A	N/A
8737708	12/2013	Hartmann et al.	N/A	N/A
8746887	12/2013	Shestak et al.	N/A	N/A
8764025	12/2013	Gao	N/A	N/A
8784450	12/2013	Moskowitz et al.	N/A	N/A
8786689	12/2013	Liu	N/A	N/A

D710545	12/2013	Wu	N/A	N/A
D710546	12/2013	Wu	N/A	N/A
8827934	12/2013	Chopra et al.	N/A	N/A
8831706	12/2013	Fu et al.	N/A	N/A
8836768	12/2013	Rafii et al.	N/A	N/A
8838199	12/2013	Simon et al.	N/A	N/A
8848977	12/2013	Bammer et al.	N/A	N/A
8855395	12/2013	Baturin et al.	N/A	N/A
8878900	12/2013	Yang et al.	N/A	N/A
8879815	12/2013	Miao et al.	N/A	N/A
8885177	12/2013	Ben-Yishai et al.	N/A	N/A
8890772	12/2013	Woo et al.	N/A	N/A
8890773	12/2013	Pederson	N/A	N/A
8890943	12/2013	Lee et al.	N/A	N/A
8897514	12/2013	Feikas et al.	N/A	N/A
8900131	12/2013	Chopra et al.	N/A	N/A
8903150	12/2013	Star-Lack et al.	N/A	N/A
8908952	12/2013	Isaacs et al.	N/A	N/A
8911358	12/2013	Koninckx et al.	N/A	N/A
8917268	12/2013	Johnsen et al.	N/A	N/A
8920776	12/2013	Gaiger et al.	N/A	N/A
8922589	12/2013	Laor	N/A	N/A
8941559	12/2014	Bar-Zeev et al.	N/A	N/A
8942455	12/2014	Chou et al.	N/A	N/A
8950877	12/2014	Northey et al.	N/A	N/A
8953246	12/2014	Koenig	N/A	N/A
8961500	12/2014	Dicorleto et al.	N/A	N/A
8965583	12/2014	Ortmaier et al.	N/A	N/A
8969829	12/2014	Wollenweber et al.	N/A	N/A
8989349	12/2014	Thomson et al.	N/A	N/A
8992580	12/2014	Bar et al.	N/A	N/A
8994729	12/2014	Nakamura	N/A	N/A
8994795	12/2014	Oh	N/A	N/A
9004711	12/2014	Gerolemou	N/A	N/A
9005211	12/2014	Brundobler et al.	N/A	N/A
9011441	12/2014	Bertagnoli et al.	N/A	N/A
9057759	12/2014	Klingenbeck et al.	N/A	N/A
9060757	12/2014	Lawson et al.	N/A	N/A
9066751	12/2014	Sasso	N/A	N/A
9081436	12/2014	Berme et al.	N/A	N/A
9084635	12/2014	Nuckley et al.	N/A	N/A
9085643	12/2014	Svanborg et al.	N/A	N/A
9087471	12/2014	Miao	N/A	N/A
9100643	12/2014	McDowall et al.	N/A	N/A
9101394	12/2014	Arata et al.	N/A	N/A
9104902	12/2014	Xu et al.	N/A	N/A
9111175	12/2014	Strommer et al.	N/A	N/A
9123155	12/2014	Cunningham et al.	N/A	N/A
9125556	12/2014	Zehavi et al.	N/A	N/A
9129054	12/2014	Nawana et al.	N/A	N/A

9129372	12/2014	Kriston et al.	N/A	N/A
9132361	12/2014	Smithwick	N/A	N/A
9135706	12/2014	Zagorchev et al.	N/A	N/A
9141873	12/2014	Takemoto	N/A	N/A
9142020	12/2014	Deguisse et al.	N/A	N/A
9149317	12/2014	Arthur et al.	N/A	N/A
9165203	12/2014	McCarthy	N/A	N/A
9165362	12/2014	Siewerdsen et al.	N/A	N/A
9179984	12/2014	Teichman et al.	N/A	N/A
D746354	12/2014	Chang	N/A	N/A
9208916	12/2014	Appleby et al.	N/A	N/A
9220573	12/2014	Kendrick et al.	N/A	N/A
9225895	12/2014	Kozinski	N/A	N/A
9232982	12/2015	Soler et al.	N/A	N/A
9235934	12/2015	Mandella et al.	N/A	N/A
9240046	12/2015	Carrell et al.	N/A	N/A
9244278	12/2015	Sugiyama et al.	N/A	N/A
9247240	12/2015	Park et al.	N/A	N/A
9259192	12/2015	Ishihara	N/A	N/A
9265572	12/2015	Fuchs et al.	N/A	N/A
9269192	12/2015	Kobayashi	N/A	N/A
9283052	12/2015	Rodriguez Ponce	N/A	N/A
9286730	12/2015	Bar-Zeev et al.	N/A	N/A
9289267	12/2015	Sauer et al.	N/A	N/A
9294222	12/2015	Proctor, Jr.	N/A	N/A
9300949	12/2015	Ahearn	N/A	N/A
9305354	12/2015	Burlon et al.	N/A	N/A
9310591	12/2015	Hua et al.	N/A	N/A
9320474	12/2015	Demri et al.	N/A	N/A
9323055	12/2015	Baillot	N/A	N/A
9330477	12/2015	Rappel	N/A	N/A
9335547	12/2015	Takano et al.	N/A	N/A
9335567	12/2015	Nakamura	N/A	N/A
9341704	12/2015	Picard et al.	N/A	N/A
9344686	12/2015	Moharir	N/A	N/A
9349066	12/2015	Koo et al.	N/A	N/A
9349520	12/2015	Demetriou et al.	N/A	N/A
9364294	12/2015	Razzaque et al.	N/A	N/A
9370332	12/2015	Paladini et al.	N/A	N/A
9373166	12/2015	Azar	N/A	N/A
9375639	12/2015	Kobayashi et al.	N/A	N/A
9378558	12/2015	Kajiwara et al.	N/A	N/A
9380287	12/2015	Nistico et al.	N/A	N/A
9387008	12/2015	Sarvestani et al.	N/A	N/A
9392129	12/2015	Simmons	N/A	N/A
9395542	12/2015	Tilleman et al.	N/A	N/A
9398936	12/2015	Razzaque et al.	N/A	N/A
9400384	12/2015	Griffith	N/A	N/A
9414041	12/2015	Ko et al.	N/A	N/A
9424611	12/2015	Kanjirathinkal et al.	N/A	N/A

9424641	12/2015	Wiemker et al.	N/A	N/A
9427286	12/2015	Siewerdsen et al.	N/A	N/A
9438894	12/2015	Park et al.	N/A	N/A
9443488	12/2015	Borenstein et al.	N/A	N/A
9453804	12/2015	Tahtali	N/A	N/A
9456878	12/2015	MacFarlane et al.	N/A	N/A
9465235	12/2015	Chang	N/A	N/A
9468373	12/2015	Larsen	N/A	N/A
9470908	12/2015	Frankel et al.	N/A	N/A
9473766	12/2015	Douglas et al.	N/A	N/A
9492222	12/2015	Singh	N/A	N/A
9495585	12/2015	Bicer et al.	N/A	N/A
9498132	12/2015	Maier-Hein et al.	N/A	N/A
9498231	12/2015	Haider et al.	N/A	N/A
9499999	12/2015	Nanqing	N/A	N/A
9507155	12/2015	Morimoto	N/A	N/A
9513495	12/2015	Waters	N/A	N/A
9521966	12/2015	Schwartz	N/A	N/A
9526443	12/2015	Berme et al.	N/A	N/A
9530382	12/2015	Simmons	N/A	N/A
9532846	12/2016	Nakamura	N/A	N/A
9532849	12/2016	Anderson et al.	N/A	N/A
9533407	12/2016	Ragner	N/A	N/A
9538962	12/2016	Hannaford et al.	N/A	N/A
9545233	12/2016	Sirpad et al.	N/A	N/A
9546779	12/2016	Rementer	N/A	N/A
9547174	12/2016	Gao et al.	N/A	N/A
9547940	12/2016	Sun et al.	N/A	N/A
9557566	12/2016	Fujimaki	N/A	N/A
9560318	12/2016	Reina et al.	N/A	N/A
9561095	12/2016	Nguyen et al.	N/A	N/A
9561446	12/2016	Brecher	N/A	N/A
9565415	12/2016	Zhang et al.	N/A	N/A
9572661	12/2016	Robin et al.	N/A	N/A
9576398	12/2016	Zehner et al.	N/A	N/A
9576556	12/2016	Simmons	N/A	N/A
9581822	12/2016	Morimoto	N/A	N/A
9610056	12/2016	Lavallee et al.	N/A	N/A
9612657	12/2016	Bertram et al.	N/A	N/A
9626936	12/2016	Bell	N/A	N/A
9629595	12/2016	Walker et al.	N/A	N/A
9633431	12/2016	Merlet	N/A	N/A
9645395	12/2016	Bolas et al.	N/A	N/A
9646423	12/2016	Sun et al.	N/A	N/A
9672597	12/2016	Amiot et al.	N/A	N/A
9672607	12/2016	Demri et al.	N/A	N/A
9672640	12/2016	Kleiner	N/A	N/A
9675306	12/2016	Morton	N/A	N/A
9675319	12/2016	Razzaque et al.	N/A	N/A
9684980	12/2016	Royalty et al.	N/A	N/A

9690119	12/2016	Garofolo et al.	N/A	N/A
RE46463	12/2016	Fienbloom et al.	N/A	N/A
9693748	12/2016	Rai et al.	N/A	N/A
9710968	12/2016	Dillavou et al.	N/A	N/A
9713502	12/2016	Finkman et al.	N/A	N/A
9724119	12/2016	Hissong et al.	N/A	N/A
9724165	12/2016	Arata et al.	N/A	N/A
9726888	12/2016	Giartosio et al.	N/A	N/A
9728006	12/2016	Varga	N/A	N/A
9729831	12/2016	Birnkrant et al.	N/A	N/A
9746739	12/2016	Alton et al.	N/A	N/A
9757034	12/2016	Desjardins et al.	N/A	N/A
9757087	12/2016	Simon et al.	N/A	N/A
9766441	12/2016	Rappel	N/A	N/A
9766459	12/2016	Alton et al.	N/A	N/A
9767608	12/2016	Lee et al.	N/A	N/A
9770203	12/2016	Berme et al.	N/A	N/A
9772102	12/2016	Ferguson	N/A	N/A
9772495	12/2016	Tam et al.	N/A	N/A
9791138	12/2016	Feinbloom et al.	N/A	N/A
9800995	12/2016	Libin et al.	N/A	N/A
9805504	12/2016	Zhang et al.	N/A	N/A
9808148	12/2016	Miller et al.	N/A	N/A
9839448	12/2016	Reckling et al.	N/A	N/A
9844413	12/2016	Daon et al.	N/A	N/A
9851080	12/2016	Wilt et al.	N/A	N/A
9858663	12/2017	Penney et al.	N/A	N/A
9861446	12/2017	Lang	N/A	N/A
9864214	12/2017	Fass	N/A	N/A
9872733	12/2017	Shoham et al.	N/A	N/A
9875544	12/2017	Rai et al.	N/A	N/A
9877642	12/2017	Duret	N/A	N/A
9885465	12/2017	Nguyen	N/A	N/A
9886552	12/2017	Dillavou et al.	N/A	N/A
9886760	12/2017	Liu et al.	N/A	N/A
9892564	12/2017	Cvetko	N/A	G06T 15/04
9898866	12/2017	Fuchs et al.	N/A	N/A
9901414	12/2017	Lively et al.	N/A	N/A
9911187	12/2017	Steinle et al.	N/A	N/A
9911236	12/2017	Bar et al.	N/A	N/A
9927611	12/2017	Rudy et al.	N/A	N/A
9928629	12/2017	Benishti et al.	N/A	N/A
9940750	12/2017	Dillavou et al.	N/A	N/A
9943374	12/2017	Merritt et al.	N/A	N/A
9947110	12/2017	Haimerl	N/A	N/A
9952664	12/2017	Border et al.	N/A	N/A
9956054	12/2017	Aguirre-Valencia	N/A	N/A
9958674	12/2017	Border	N/A	N/A
9959620	12/2017	Merlet	N/A	N/A

9959629	12/2017	Dillavou et al.	N/A	N/A
9965681	12/2017	Border et al.	N/A	N/A
9968297	12/2017	Connor	N/A	N/A
9980780	12/2017	Lang	N/A	N/A
9986228	12/2017	Woods	N/A	N/A
D824523	12/2017	Paoli et al.	N/A	N/A
10010379	12/2017	Gibby et al.	N/A	N/A
10013531	12/2017	Richards et al.	N/A	N/A
10015243	12/2017	Kazerani et al.	N/A	N/A
10016243	12/2017	Esterberg	N/A	N/A
10022064	12/2017	Kim et al.	N/A	N/A
10022065	12/2017	Ben-Yishai et al.	N/A	N/A
10022104	12/2017	Sell et al.	N/A	N/A
10023615	12/2017	Bonny	N/A	N/A
10026015	12/2017	Cavusoglu et al.	N/A	N/A
10034713	12/2017	Yang et al.	N/A	N/A
10042167	12/2017	McDowall et al.	N/A	N/A
10046165	12/2017	Frewin et al.	N/A	N/A
10055838	12/2017	Elenbaas et al.	N/A	N/A
10066816	12/2017	Chang	N/A	N/A
10067359	12/2017	Ushakov	N/A	N/A
10073515	12/2017	Awdeh	N/A	N/A
10080616	12/2017	Wilkinson et al.	N/A	N/A
10082680	12/2017	Chung	N/A	N/A
10085709	12/2017	Lavallee et al.	N/A	N/A
10105187	12/2017	Corndorf et al.	N/A	N/A
10107483	12/2017	Oren	N/A	N/A
10108833	12/2017	Hong et al.	N/A	N/A
10123840	12/2017	Dorman	N/A	N/A
10130378	12/2017	Bryan	N/A	N/A
10132483	12/2017	Feinbloom et al.	N/A	N/A
10134166	12/2017	Benishti et al.	N/A	N/A
10134194	12/2017	Kepner et al.	N/A	N/A
10139652	12/2017	Windham	N/A	N/A
10139920	12/2017	Isaacs et al.	N/A	N/A
10142496	12/2017	Rao et al.	N/A	N/A
10151928	12/2017	Ushakov	N/A	N/A
10154239	12/2017	Casas	N/A	N/A
10159530	12/2017	Lang	N/A	N/A
10163207	12/2017	Merlet	N/A	N/A
10166079	12/2018	McLachlin et al.	N/A	N/A
10175507	12/2018	Nakamura	N/A	N/A
10175753	12/2018	Boesen	N/A	N/A
10181361	12/2018	Dillavou et al.	N/A	N/A
10186055	12/2018	Takahashi et al.	N/A	N/A
10188672	12/2018	Wagner	N/A	N/A
10194131	12/2018	Casas	N/A	N/A
10194990	12/2018	Amanatullah et al.	N/A	N/A
10194993	12/2018	Roger et al.	N/A	N/A
10195076	12/2018	Fateh	N/A	N/A

10197803	12/2018	Badiali et al.	N/A	N/A
10197816	12/2018	Waisman et al.	N/A	N/A
10207315	12/2018	Appleby et al.	N/A	N/A
10212517	12/2018	Beltran et al.	N/A	N/A
10230719	12/2018	Vaughn et al.	N/A	N/A
10231893	12/2018	Lei et al.	N/A	N/A
10235606	12/2018	Miao et al.	N/A	N/A
10240769	12/2018	Braganca et al.	N/A	N/A
10247965	12/2018	Ton	N/A	N/A
10251724	12/2018	McLachlin et al.	N/A	N/A
10261324	12/2018	Chuang et al.	N/A	N/A
10262424	12/2018	Ketcha et al.	N/A	N/A
10274731	12/2018	Maimone	N/A	N/A
10278777	12/2018	Lang	N/A	N/A
10292768	12/2018	Lang	N/A	N/A
10296805	12/2018	Yang et al.	N/A	N/A
10319154	12/2018	Chakravarthula et al.	N/A	N/A
10326975	12/2018	Casas	N/A	N/A
10332267	12/2018	Rai et al.	N/A	N/A
10339719	12/2018	Jagga et al.	N/A	N/A
10352543	12/2018	Braganca et al.	N/A	N/A
10357146	12/2018	Fiebel et al.	N/A	N/A
10357574	12/2018	Hilderbrand et al.	N/A	N/A
10366489	12/2018	Boettger et al.	N/A	N/A
10368947	12/2018	Lang	N/A	N/A
10368948	12/2018	Tripathi	N/A	N/A
10382748	12/2018	Benishti et al.	N/A	N/A
10383654	12/2018	Yilmaz et al.	N/A	N/A
10386645	12/2018	Abou Shousha	N/A	N/A
10388076	12/2018	Bar-Zeev et al.	N/A	N/A
10398514	12/2018	Ryan et al.	N/A	N/A
10401657	12/2018	Jiang et al.	N/A	N/A
10405825	12/2018	Rai et al.	N/A	N/A
10405927	12/2018	Lang	N/A	N/A
10413752	12/2018	Berlinger et al.	N/A	N/A
10419655	12/2018	Sivan	N/A	N/A
10420626	12/2018	Tokuda et al.	N/A	N/A
10420813	12/2018	Newell-Rogers et al.	N/A	N/A
10424115	12/2018	Ellerbrock	N/A	N/A
D862469	12/2018	Sadot et al.	N/A	N/A
10426554	12/2018	Siewerdsen et al.	N/A	N/A
10429675	12/2018	Greget	N/A	N/A
10431008	12/2018	Djajadiningrat et al.	N/A	N/A
10433814	12/2018	Razzaque et al.	N/A	N/A
10434335	12/2018	Takahashi et al.	N/A	N/A
10441236	12/2018	Bar-Tal et al.	N/A	N/A
10444514	12/2018	Abou Shousha et al.	N/A	N/A
10447947	12/2018	Liu	N/A	N/A
10448003	12/2018	Grafenberg	N/A	N/A
10449040	12/2018	Lashinski et al.	N/A	N/A

10453187	12/2018	Peterson et al.	N/A	N/A
10463434	12/2018	Siegler et al.	N/A	N/A
10465892	12/2018	Feinbloom et al.	N/A	N/A
10466487	12/2018	Blum et al.	N/A	N/A
10470732	12/2018	Baumgart et al.	N/A	N/A
10473314	12/2018	Braganca et al.	N/A	N/A
10485989	12/2018	Jordan et al.	N/A	N/A
10488663	12/2018	Choi	N/A	N/A
D869772	12/2018	Gand	N/A	N/A
D870977	12/2018	Berggren et al.	N/A	N/A
10492755	12/2018	Lin et al.	N/A	N/A
10499997	12/2018	Weinstein et al.	N/A	N/A
10502363	12/2018	Edwards et al.	N/A	N/A
10504231	12/2018	Fiala	N/A	N/A
10507066	12/2018	Dimaio et al.	N/A	N/A
10511822	12/2018	Casas	N/A	N/A
10517544	12/2018	Taguchi et al.	N/A	N/A
10537395	12/2019	Perez	N/A	N/A
10540780	12/2019	Cousins et al.	N/A	N/A
10543485	12/2019	Ismagilov et al.	N/A	N/A
10546423	12/2019	Jones et al.	N/A	N/A
10548557	12/2019	Lim et al.	N/A	N/A
10555775	12/2019	Hoffman et al.	N/A	N/A
10568535	12/2019	Roberts et al.	N/A	N/A
10571696	12/2019	Urey et al.	N/A	N/A
10571716	12/2019	Chapiro	N/A	N/A
10573086	12/2019	Bar-Zeev et al.	N/A	N/A
10573087	12/2019	Gallop et al.	N/A	N/A
10577630	12/2019	Zhang et al.	N/A	N/A
10586400	12/2019	Douglas	N/A	N/A
10591737	12/2019	Yildiz et al.	N/A	N/A
10592748	12/2019	Cousins et al.	N/A	N/A
10594998	12/2019	Casas	N/A	N/A
10595716	12/2019	Nazareth et al.	N/A	N/A
10601950	12/2019	Devam et al.	N/A	N/A
10602114	12/2019	Casas	N/A	N/A
10603113	12/2019	Lang	N/A	N/A
10603133	12/2019	Wang et al.	N/A	N/A
10606085	12/2019	Toyama	N/A	N/A
10610172	12/2019	Hummel et al.	N/A	N/A
10610179	12/2019	Altmann	N/A	N/A
10613352	12/2019	Knoll	N/A	N/A
10617566	12/2019	Esmonde	N/A	N/A
10620460	12/2019	Carabin	N/A	N/A
10621738	12/2019	Miao et al.	N/A	N/A
10625099	12/2019	Takahashi et al.	N/A	N/A
10626473	12/2019	Mariani et al.	N/A	N/A
10631905	12/2019	Asfora et al.	N/A	N/A
10631907	12/2019	Zucker et al.	N/A	N/A
10634331	12/2019	Feinbloom et al.	N/A	N/A

10634921	12/2019	Blum et al.	N/A	N/A
10638080	12/2019	Ovchinnikov et al.	N/A	N/A
10646285	12/2019	Siemionow et al.	N/A	N/A
10650513	12/2019	Penney et al.	N/A	N/A
10650594	12/2019	Jones et al.	N/A	N/A
10652525	12/2019	Woods	N/A	N/A
10653495	12/2019	Gregerson	N/A	A61B 6/461
10660715	12/2019	Dozeman	N/A	N/A
10663738	12/2019	Carlvik et al.	N/A	N/A
10665033	12/2019	Bar-Zeev et al.	N/A	N/A
10670937	12/2019	Alton et al.	N/A	N/A
10672145	12/2019	Albiol et al.	N/A	N/A
10682112	12/2019	Pizaine et al.	N/A	N/A
10682767	12/2019	Grafenberg et al.	N/A	N/A
10687901	12/2019	Thomas	N/A	N/A
10691397	12/2019	Clements	N/A	N/A
10702713	12/2019	Mori et al.	N/A	N/A
10706540	12/2019	Merlet	N/A	N/A
10709398	12/2019	Schweizer	N/A	N/A
10713801	12/2019	Jordan et al.	N/A	N/A
10716643	12/2019	Justin et al.	N/A	N/A
10722733	12/2019	Takahashi	N/A	N/A
10725535	12/2019	Yu	N/A	N/A
10731832	12/2019	Koo	N/A	N/A
10732721	12/2019	Clements	N/A	N/A
10742949	12/2019	Casas	N/A	N/A
10743939	12/2019	Lang	N/A	N/A
10743943	12/2019	Razeto et al.	N/A	N/A
10747315	12/2019	Tungare et al.	N/A	N/A
10748319	12/2019	Tao et al.	N/A	N/A
10758315	12/2019	Johnson et al.	N/A	N/A
10777094	12/2019	Rao et al.	N/A	N/A
10777315	12/2019	Zehavi et al.	N/A	N/A
10781482	12/2019	Gubatayao et al.	N/A	N/A
10792110	12/2019	Leung et al.	N/A	N/A
10799145	12/2019	West et al.	N/A	N/A
10799296	12/2019	Lang	N/A	N/A
10799298	12/2019	Crawford et al.	N/A	N/A
10799316	12/2019	Sela et al.	N/A	N/A
10810799	12/2019	Tepper et al.	N/A	N/A
10818019	12/2019	Piat et al.	N/A	N/A
10818101	12/2019	Gallop et al.	N/A	N/A
10818199	12/2019	Buras et al.	N/A	N/A
10825563	12/2019	Gibby et al.	N/A	N/A
10827164	12/2019	Perreault et al.	N/A	N/A
10831943	12/2019	Santarone et al.	N/A	N/A
10835296	12/2019	Elimelech et al.	N/A	N/A
10838206	12/2019	Fortin-Deschnes et al.	N/A	N/A
10839629	12/2019	Jones et al.	N/A	N/A

10839956	12/2019	Beydoun et al.	N/A	N/A
10841556	12/2019	Casas	N/A	N/A
10842002	12/2019	Chang	N/A	N/A
10842461	12/2019	Johnson et al.	N/A	N/A
10849691	12/2019	Zucker et al.	N/A	N/A
10849693	12/2019	Lang	N/A	N/A
10849710	12/2019	Liu	N/A	N/A
10861236	12/2019	Geri et al.	N/A	N/A
10865220	12/2019	Ebetino et al.	N/A	N/A
10869517	12/2019	Halpern	N/A	N/A
10869727	12/2019	Yanof et al.	N/A	N/A
10872472	12/2019	Watola et al.	N/A	N/A
10877262	12/2019	Luxembourg	N/A	N/A
10877296	12/2019	Lindsey et al.	N/A	N/A
10878639	12/2019	Douglas et al.	N/A	N/A
10893260	12/2020	Trail et al.	N/A	N/A
10895742	12/2020	Schneider et al.	N/A	N/A
10895743	12/2020	Dausmann	N/A	N/A
10895906	12/2020	West et al.	N/A	N/A
10898151	12/2020	Harding et al.	N/A	N/A
10908420	12/2020	Lee et al.	N/A	N/A
10921595	12/2020	Rakshit et al.	N/A	N/A
10921613	12/2020	Gupta et al.	N/A	N/A
10928321	12/2020	Rawle	N/A	N/A
10928638	12/2020	Ninan et al.	N/A	N/A
10929670	12/2020	Troy et al.	N/A	N/A
10935815	12/2020	Castaeda	N/A	N/A
10935816	12/2020	Ban et al.	N/A	N/A
10936537	12/2020	Huston	N/A	N/A
10939973	12/2020	Dimaio et al.	N/A	N/A
10939977	12/2020	Messinger et al.	N/A	N/A
10941933	12/2020	Ferguson	N/A	N/A
10946108	12/2020	Zhang et al.	N/A	N/A
10950338	12/2020	Douglas	N/A	N/A
10951872	12/2020	Casas	N/A	N/A
10964095	12/2020	Douglas	N/A	N/A
10964124	12/2020	Douglas	N/A	N/A
10966768	12/2020	Poulos	N/A	N/A
10969587	12/2020	McDowall et al.	N/A	N/A
10993754	12/2020	Kuntz et al.	N/A	N/A
11000335	12/2020	Dorman	N/A	N/A
11002994	12/2020	Jiang et al.	N/A	N/A
11006093	12/2020	Hegyi	N/A	N/A
11013550	12/2020	Rioux et al.	N/A	N/A
11013560	12/2020	Lang	N/A	N/A
11013562	12/2020	Marti et al.	N/A	N/A
11013573	12/2020	Chang	N/A	N/A
11013900	12/2020	Malek et al.	N/A	N/A
11016302	12/2020	Freeman et al.	N/A	N/A
11019988	12/2020	Fiebel et al.	N/A	N/A

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11030809	12/2020	Wang	N/A	N/A
11041173	12/2020	Zhang et al.	N/A	N/A
11045663	12/2020	Mori et al.	N/A	N/A
11049293	12/2020	Chae et al.	N/A	N/A
11049476	12/2020	Fuchs et al.	N/A	N/A
11050990	12/2020	Casas	N/A	N/A
11057505	12/2020	Dharmatilleke	N/A	N/A
11058390	12/2020	Douglas	N/A	N/A
11061257	12/2020	Hakim	N/A	N/A
11064904	12/2020	Kay et al.	N/A	N/A
11065062	12/2020	Frushour et al.	N/A	N/A
11067387	12/2020	Marell et al.	N/A	N/A
11071497	12/2020	Hallack et al.	N/A	N/A
11079596	12/2020	Hua et al.	N/A	N/A
11087039	12/2020	Duff et al.	N/A	N/A
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11097129	12/2020	Sakata et al.	N/A	N/A
11099376	12/2020	Steier et al.	N/A	N/A
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11109762	12/2020	Steier et al.	N/A	N/A
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11123604	12/2020	Fung	N/A	N/A
11129562	12/2020	Roberts et al.	N/A	N/A
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11135015	12/2020	Crawford et al.	N/A	N/A
11135016	12/2020	Frielinghaus et al.	N/A	N/A
11137610	12/2020	Kessler et al.	N/A	N/A
11141221	12/2020	Hobeika et al.	N/A	N/A
11153549	12/2020	Casas	N/A	N/A
11153555	12/2020	Healy et al.	N/A	N/A
11163176	12/2020	Karafin et al.	N/A	N/A
11164324	12/2020	Liu et al.	N/A	N/A
11166006	12/2020	Hegyi	N/A	N/A
11169380	12/2020	Manly et al.	N/A	N/A
11172990	12/2020	Lang	N/A	N/A
11179136	12/2020	Kohli et al.	N/A	N/A
11180557	12/2020	Noelle	N/A	N/A
11181747	12/2020	Kessler et al.	N/A	N/A
11185891	12/2020	Cousins et al.	N/A	N/A
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11202682	12/2020	Staunton et al.	N/A	N/A
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11227417	12/2021	Berlinger et al.	N/A	N/A

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11244508	12/2021	Kazanzides et al.	N/A	N/A
11253216	12/2021	Crawford et al.	N/A	N/A
11253323	12/2021	Hughes et al.	N/A	N/A
11257190	12/2021	Mao et al.	N/A	N/A
11257241	12/2021	Tao	N/A	N/A
11263772	12/2021	Siemionow et al.	N/A	N/A
11269401	12/2021	West et al.	N/A	N/A
11272151	12/2021	Casas	N/A	N/A
11278359	12/2021	Siemionow et al.	N/A	N/A
11278413	12/2021	Lang	N/A	N/A
11280480	12/2021	Wilt et al.	N/A	N/A
11284846	12/2021	Graumann et al.	N/A	N/A
11291521	12/2021	Im	N/A	N/A
11294167	12/2021	Ishimoda	N/A	N/A
11297285	12/2021	Pierce	N/A	N/A
11300252	12/2021	Nguyen	N/A	N/A
11300790	12/2021	Cheng et al.	N/A	N/A
11304621	12/2021	Merschon et al.	N/A	N/A
11304759	12/2021	Kovtun et al.	N/A	N/A
11307402	12/2021	Steier et al.	N/A	N/A
11308663	12/2021	Alhrishy et al.	N/A	N/A
11311341	12/2021	Lang	N/A	N/A
11317973	12/2021	Calloway et al.	N/A	N/A
11337763	12/2021	Choi	N/A	N/A
11348257	12/2021	Lang	N/A	N/A
11350072	12/2021	Quiles Casas	N/A	N/A
11350965	12/2021	Yilmaz et al.	N/A	N/A
11351006	12/2021	Aferzon et al.	N/A	N/A
11354813	12/2021	Piat et al.	N/A	N/A
11360315	12/2021	Tu et al.	N/A	N/A
11373342	12/2021	Stafford et al.	N/A	N/A
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11382700	12/2021	Calloway et al.	N/A	N/A
11382712	12/2021	Elimelech et al.	N/A	N/A
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11393229	12/2021	Zhou et al.	N/A	N/A
11399895	12/2021	Soper et al.	N/A	N/A
11402524	12/2021	Song et al.	N/A	N/A
11406338	12/2021	Tolkowsky	N/A	N/A
11412202	12/2021	Hegyi	N/A	N/A
11423554	12/2021	Borsdorf et al.	N/A	N/A
11430203	12/2021	Navab et al.	N/A	N/A
11432828	12/2021	Lang	N/A	N/A
11432931	12/2021	Lang	N/A	N/A
11443428	12/2021	Petersen et al.	N/A	N/A
11443431	12/2021	Flossmann et al.	N/A	N/A
11452568	12/2021	Lang	N/A	N/A

11452570	12/2021	Tolkowsky	N/A	N/A
11460915	12/2021	Frielinghaus et al.	N/A	N/A
11461936	12/2021	Freeman et al.	N/A	N/A
11461983	12/2021	Jones et al.	N/A	N/A
11464580	12/2021	Kemp et al.	N/A	N/A
11464581	12/2021	Calloway	N/A	N/A
11475625	12/2021	Douglas	N/A	N/A
11478214	12/2021	Siewerdsen et al.	N/A	N/A
11483532	12/2021	Quiles Casas	N/A	N/A
11488021	12/2021	Sun et al.	N/A	N/A
11490986	12/2021	BEn-Yishai	N/A	N/A
11510750	12/2021	Dulin et al.	N/A	N/A
11513358	12/2021	McDowall et al.	N/A	N/A
11527002	12/2021	Govari	N/A	N/A
11528393	12/2021	Garofolo et al.	N/A	N/A
11544031	12/2022	Harviainen	N/A	N/A
11573420	12/2022	Sarma et al.	N/A	N/A
11589927	12/2022	Oezbek et al.	N/A	N/A
11627924	12/2022	Alexandroni et al.	N/A	N/A
11644675	12/2022	Manly et al.	N/A	N/A
11648016	12/2022	Hathaway et al.	N/A	N/A
11651499	12/2022	Wang et al.	N/A	N/A
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11666458	12/2022	Kim et al.	N/A	N/A
11669984	12/2022	Siewerdsen et al.	N/A	N/A
11686947	12/2022	Loyola et al.	N/A	N/A
11699236	12/2022	Avital et al.	N/A	N/A
11712582	12/2022	Miyazaki et al.	N/A	N/A
11715210	12/2022	Haslam et al.	N/A	N/A
11719941	12/2022	Russell	N/A	N/A
11730389	12/2022	Farshad et al.	N/A	N/A
11733516	12/2022	Edwin et al.	N/A	N/A
11734901	12/2022	Jones et al.	N/A	N/A
11744657	12/2022	Leboeuf et al.	N/A	N/A
11750794	12/2022	Benishti et al.	N/A	N/A
11766296	12/2022	Wolf et al.	N/A	N/A
11798178	12/2022	Merlet	N/A	N/A
11801097	12/2022	Crawford et al.	N/A	N/A
11801115	12/2022	Elimelech et al.	N/A	N/A
11808943	12/2022	Robaina et al.	N/A	N/A
11815683	12/2022	Sears et al.	N/A	N/A
11826111	12/2022	Mahfouz	N/A	N/A
11832886	12/2022	Dorman	N/A	N/A
11838493	12/2022	Healy et al.	N/A	N/A
11839433	12/2022	Schaewe et al.	N/A	N/A
11839501	12/2022	Takahashi et al.	N/A	N/A
11864934	12/2023	Junio et al.	N/A	N/A
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11896445	12/2023	Gera et al.	N/A	N/A

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11922631	12/2023	Haslam et al.	N/A	N/A
11941814	12/2023	Crawford et al.	N/A	N/A
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11948265	12/2023	Gibby et al.	N/A	N/A
11950968	12/2023	Wiggermann	N/A	N/A
11957420	12/2023	Lang	N/A	N/A
11961193	12/2023	Pelzl et al.	N/A	N/A
11963723	12/2023	Vilsmeier et al.	N/A	N/A
11972582	12/2023	Yan et al.	N/A	N/A
11974819	12/2023	Finley et al.	N/A	N/A
11974887	12/2023	Elimelech et al.	N/A	N/A
11977232	12/2023	Wu et al.	N/A	N/A
11980429	12/2023	Wolf et al.	N/A	N/A
11980506	12/2023	Wolf et al.	N/A	N/A
11980507	12/2023	Elimelech et al.	N/A	N/A
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11983824	12/2023	Avisar et al.	N/A	N/A
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12010285	12/2023	Quiles Casas	N/A	N/A
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12019314	12/2023	Steines et al.	N/A	N/A
12026897	12/2023	Frantz et al.	N/A	N/A
12033322	12/2023	Laaksonen et al.	N/A	N/A
12044856	12/2023	Gera et al.	N/A	N/A
12044858	12/2023	Gera et al.	N/A	N/A
12053247	12/2023	Chiou	N/A	N/A
12056830	12/2023	Cvetko et al.	N/A	N/A
12059281	12/2023	Weingarten et al.	N/A	N/A
12063338	12/2023	Quiles Casas	N/A	N/A
12063345	12/2023	Benishti et al.	N/A	N/A
12069233	12/2023	Benishti et al.	N/A	N/A
12076158	12/2023	Geiger et al.	N/A	N/A
12076196	12/2023	Elimelech et al.	N/A	N/A
12079385	12/2023	Ben-Yishai et al.	N/A	N/A
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12114933	12/2023	Seo et al.	N/A	N/A
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12178666	12/2023	Wolf et al.	N/A	N/A
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12239385	12/2024	Wolf et al.	N/A	N/A

2002/0082498	12/2001	Wendt et al.	N/A	N/A
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2003/0117393	12/2002	Sauer et al.	N/A	N/A
2003/0130576	12/2002	Seeley et al.	N/A	N/A
2003/0156144	12/2002	Morita	N/A	N/A
2003/0210812	12/2002	Khamene et al.	N/A	N/A
2003/0225329	12/2002	Rossner et al.	N/A	N/A
2004/0019263	12/2003	Jutras et al.	N/A	N/A
2004/0030237	12/2003	Lee et al.	N/A	N/A
2004/0138556	12/2003	Cosman	N/A	N/A
2004/0152955	12/2003	McGinley et al.	N/A	N/A
2004/0171930	12/2003	Grimm et al.	N/A	N/A
2004/0238732	12/2003	State et al.	N/A	N/A
2005/0017972	12/2004	Poole et al.	N/A	N/A
2005/0024586	12/2004	Teiwes et al.	N/A	N/A
2005/0119639	12/2004	McCombs et al.	N/A	N/A
2005/0154296	12/2004	Lechner et al.	N/A	N/A
2005/0203367	12/2004	Ahmed et al.	N/A	N/A
2005/0203380	12/2004	Sauer et al.	N/A	N/A
2005/0215879	12/2004	Chuanggui	N/A	N/A
2005/0267358	12/2004	Tuma et al.	N/A	N/A
2006/0072124	12/2005	Smetak et al.	N/A	N/A
2006/0134198	12/2005	Tawa et al.	N/A	N/A
2006/0147100	12/2005	Fitzpatrick	N/A	N/A
2006/0176242	12/2005	Jaramaz et al.	N/A	N/A
2006/0241760	12/2005	Randall et al.	N/A	N/A
2007/0018975	12/2006	Chuanggui et al.	N/A	N/A
2007/0058261	12/2006	Sugihara et al.	N/A	N/A
2007/0100325	12/2006	Jutras et al.	N/A	N/A
2007/0183041	12/2006	McCloy et al.	N/A	N/A
2007/0233371	12/2006	Stoschek et al.	N/A	N/A
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2008/0159612	12/2007	Fu et al.	N/A	N/A
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2008/0221625	12/2007	Hufner et al.	N/A	N/A
2008/0253527	12/2007	Boyden et al.	N/A	N/A
2008/0262812	12/2007	Arata et al.	N/A	N/A
2008/0287728	12/2007	Mostafavi et al.	N/A	N/A
2009/0005961	12/2008	Grabowski et al.	N/A	N/A
2009/0018437	12/2008	Cooke	N/A	N/A
2009/0024127	12/2008	Lechner et al.	N/A	N/A
2009/0036902	12/2008	DiMaio et al.	N/A	N/A
2009/0062869	12/2008	Claverie et al.	N/A	N/A
2009/0099445	12/2008	Burger	N/A	N/A
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2009/0285366	12/2008	Essenreiter et al.	N/A	N/A
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2010/0094308	12/2009	Tatsumi et al.	N/A	N/A
2010/0106010	12/2009	Rubner et al.	N/A	N/A
2010/0114110	12/2009	Taft et al.	N/A	N/A
2010/0138939	12/2009	Bentzon et al.	N/A	N/A
2010/0149073	12/2009	Chaum et al.	N/A	N/A
2010/0172567	12/2009	Prokoski	N/A	N/A
2010/0210939	12/2009	Hartmann et al.	N/A	N/A
2010/0266220	12/2009	Zagorchev et al.	N/A	N/A
2010/0274124	12/2009	Jascob et al.	N/A	N/A
2011/0004259	12/2010	Stallings et al.	N/A	N/A
2011/0098553	12/2010	Robbins et al.	N/A	N/A
2011/0105895	12/2010	Kornblau et al.	N/A	N/A
2011/0125159	12/2010	Hanson et al.	N/A	N/A
2011/0125160	12/2010	Bagga et al.	N/A	N/A
2011/0216060	12/2010	Weising et al.	N/A	N/A
2011/0245625	12/2010	Trovato et al.	N/A	N/A
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2011/0254922	12/2010	Schaerer et al.	N/A	N/A
2011/0306873	12/2010	Shenai et al.	N/A	N/A
2012/0014608	12/2011	Watanabe	N/A	N/A
2012/0068913	12/2011	Bar-Zeev et al.	N/A	N/A
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2012/0182605	12/2011	Hall et al.	N/A	N/A
2012/0201421	12/2011	Hartmann et al.	N/A	N/A
2012/0216411	12/2011	Wevers et al.	N/A	N/A
2012/0224260	12/2011	Healy et al.	N/A	N/A
2012/0238609	12/2011	Srivastava et al.	N/A	N/A
2012/0245645	12/2011	Hanson et al.	N/A	N/A
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2021/0161614	12/2020	Elimelech et al.	N/A	N/A
2021/0162287	12/2020	Xing et al.	N/A	N/A
2021/0165207	12/2020	Peyman	N/A	N/A
2021/0169504	12/2020	Brown	N/A	N/A
2021/0169578	12/2020	Calloway et al.	N/A	N/A
2021/0169581	12/2020	Calloway et al.	N/A	N/A
2021/0169605	12/2020	Calloway et al.	N/A	N/A
2021/0186647	12/2020	Elimelech et al.	N/A	N/A
2021/0196404	12/2020	Wang	N/A	N/A
2021/0211640	12/2020	Bristol et al.	N/A	N/A
2021/0223577	12/2020	Zhang et al.	N/A	N/A
2021/0225006	12/2020	Grady et al.	N/A	N/A
2021/0227791	12/2020	De et al.	N/A	N/A
2021/0231301	12/2020	Hikmet et al.	N/A	N/A
2021/0235061	12/2020	Hegyi	N/A	N/A
2021/0248822	12/2020	Choi et al.	N/A	N/A
2021/0274281	12/2020	Zhang et al.	N/A	N/A
2021/0278675	12/2020	Klug et al.	N/A	N/A
2021/0282887	12/2020	Wiggermann	N/A	N/A
2021/0290046	12/2020	Nazareth et al.	N/A	N/A
2021/0290336	12/2020	Wang	N/A	N/A
2021/0290394	12/2020	Mahfouz	N/A	N/A
2021/0295108	12/2020	Bar	N/A	N/A
2021/0295512	12/2020	Knoplioch et al.	N/A	N/A
2021/0298795	12/2020	Bowling et al.	N/A	N/A
2021/0298835	12/2020	Wang	N/A	N/A
2021/0306599	12/2020	Pierce	N/A	N/A
2021/0311322	12/2020	Belanger et al.	N/A	N/A
2021/0314502	12/2020	Liu	N/A	N/A
2021/0315636	12/2020	Akbarian et al.	N/A	N/A
2021/0315662	12/2020	Freeman et al.	N/A	N/A
2021/0325684	12/2020	Ninan et al.	N/A	N/A
2021/0332447	12/2020	Lubelski et al.	N/A	N/A

2021/0333561	12/2020	Oh et al.	N/A	N/A
2021/0341739	12/2020	Cakmakci et al.	N/A	N/A
2021/0341740	12/2020	Cakmakci et al.	N/A	N/A
2021/0346115	12/2020	Dulin et al.	N/A	N/A
2021/0349677	12/2020	Baldev et al.	N/A	N/A
2021/0364802	12/2020	Uchiyama et al.	N/A	N/A
2021/0369226	12/2020	Siemionow et al.	N/A	N/A
2021/0371413	12/2020	Thurston et al.	N/A	N/A
2021/0373333	12/2020	Moon	N/A	N/A
2021/0373344	12/2020	Loyola et al.	N/A	N/A
2021/0378757	12/2020	Bay et al.	N/A	N/A
2021/0382310	12/2020	Freeman et al.	N/A	N/A
2021/0386482	12/2020	Gera et al.	N/A	N/A
2021/0389590	12/2020	Freeman et al.	N/A	N/A
2021/0400247	12/2020	Casas	N/A	N/A
2021/0401533	12/2020	Im	N/A	N/A
2021/0402255	12/2020	Fung	N/A	N/A
2021/0405369	12/2020	King	N/A	N/A
2022/0003992	12/2021	Ahn	N/A	N/A
2022/0007006	12/2021	Healy et al.	N/A	N/A
2022/0008135	12/2021	Frielinghaus et al.	N/A	N/A
2022/0038675	12/2021	Hegyi	N/A	N/A
2022/0039873	12/2021	Harris	N/A	N/A
2022/0051484	12/2021	Jones et al.	N/A	N/A
2022/0054199	12/2021	Sivaprakasam et al.	N/A	N/A
2022/0061921	12/2021	Crawford et al.	N/A	N/A
2022/0071712	12/2021	Wolf et al.	N/A	N/A
2022/0079675	12/2021	Lang	N/A	N/A
2022/0087746	12/2021	Lang	N/A	N/A
2022/0113810	12/2021	Isaacs et al.	N/A	N/A
2022/0117669	12/2021	Nikou et al.	N/A	N/A
2022/0121041	12/2021	Hakim	N/A	N/A
2022/0125496	12/2021	Lpez et al.	N/A	N/A
2022/0133484	12/2021	Lang	N/A	N/A
2022/0142730	12/2021	Wolf et al.	N/A	N/A
2022/0155861	12/2021	Myung et al.	N/A	N/A
2022/0159227	12/2021	Quiles Casas	N/A	N/A
2022/0179209	12/2021	Cherukuri	N/A	N/A
2022/0192776	12/2021	Gibby et al.	N/A	N/A
2022/0193453	12/2021	Miyazaki et al.	N/A	N/A
2022/0201274	12/2021	Achilefu et al.	N/A	N/A
2022/0245400	12/2021	Siemionow et al.	N/A	N/A
2022/0245821	12/2021	Ouzounis	N/A	N/A
2022/0257206	12/2021	Hartley et al.	N/A	N/A
2022/0269077	12/2021	Adema et al.	N/A	N/A
2022/0270263	12/2021	Junio	N/A	N/A
2022/0287676	12/2021	Steines et al.	N/A	N/A
2022/0292786	12/2021	Pelzl et al.	N/A	N/A
2022/0295033	12/2021	Quiles Casas	N/A	N/A
2022/0296315	12/2021	Sokhanvar et al.	N/A	N/A

2022/0304768	12/2021	Elimelech et al.	N/A	N/A
2022/0351385	12/2021	Finley et al.	N/A	N/A
2022/0353487	12/2021	Hegyi	N/A	N/A
2022/0358759	12/2021	Cork et al.	N/A	N/A
2022/0370152	12/2021	Lavallee et al.	N/A	N/A
2022/0387130	12/2021	Spaas et al.	N/A	N/A
2022/0392085	12/2021	Finley et al.	N/A	N/A
2022/0397750	12/2021	Zhou et al.	N/A	N/A
2022/0398752	12/2021	Yoon et al.	N/A	N/A
2022/0398755	12/2021	Herrmann	N/A	N/A
2022/0405935	12/2021	Flossmann et al.	N/A	N/A
2023/0004013	12/2022	McCracken et al.	N/A	N/A
2023/0009793	12/2022	Gera et al.	N/A	N/A
2023/0025480	12/2022	Kemp et al.	N/A	N/A
2023/0027801	12/2022	Qian et al.	N/A	N/A
2023/0032731	12/2022	Hrndler et al.	N/A	N/A
2023/0034189	12/2022	Gera et al.	N/A	N/A
2023/0050636	12/2022	Yanof et al.	N/A	N/A
2023/0053120	12/2022	Jamali et al.	N/A	N/A
2023/0073041	12/2022	Samadani et al.	N/A	N/A
2023/0085387	12/2022	Jones et al.	N/A	N/A
2023/0087783	12/2022	Dulin et al.	N/A	N/A
2023/0100078	12/2022	Toporek et al.	N/A	N/A
2023/0123621	12/2022	Joshi et al.	N/A	N/A
2023/0126207	12/2022	Wang	N/A	N/A
2023/0129056	12/2022	Hemingway et al.	N/A	N/A
2023/0131515	12/2022	Oezbek et al.	N/A	N/A
2023/0149083	12/2022	Lin et al.	N/A	N/A
2023/0162493	12/2022	Worrell et al.	N/A	N/A
2023/0165640	12/2022	Dulin et al.	N/A	N/A
2023/0169659	12/2022	Chen et al.	N/A	N/A
2023/0196582	12/2022	Grady et al.	N/A	N/A
2023/0200917	12/2022	Calloway et al.	N/A	N/A
2023/0236426	12/2022	Manly et al.	N/A	N/A
2023/0236427	12/2022	Jiannyuh	N/A	N/A
2023/0245784	12/2022	Crawford et al.	N/A	N/A
2023/0260142	12/2022	Chatterjee et al.	N/A	N/A
2023/0290037	12/2022	Tasse	345/420	G06T 15/04
2023/0295302	12/2022	Bhagavatheeswaran	424/143.1	C07K 16/2818
2023/0306590	12/2022	Jazdyk et al.	N/A	N/A
2023/0316550	12/2022	Hiasa	N/A	N/A
2023/0326011	12/2022	Cutforth et al.	N/A	N/A
2023/0326027	12/2022	Wahrenberg	N/A	N/A
2023/0329799	12/2022	Gera et al.	N/A	N/A
2023/0329801	12/2022	Elimelech et al.	N/A	N/A
2023/0334664	12/2022	Lu et al.	N/A	N/A
2023/0335261	12/2022	Reicher et al.	N/A	N/A
2023/0359043	12/2022	Russell	N/A	N/A

2023/0363832	12/2022	Mosadegh et al.	N/A	N/A
2023/0371984	12/2022	Leuthardt et al.	N/A	N/A
2023/0372053	12/2022	Elimelech et al.	N/A	N/A
2023/0372054	12/2022	Elimelech et al.	N/A	N/A
2023/0377171	12/2022	Hasler et al.	N/A	N/A
2023/0377175	12/2022	Seok	N/A	N/A
2023/0379448	12/2022	Benishti et al.	N/A	N/A
2023/0379449	12/2022	Benishti et al.	N/A	N/A
2023/0386022	12/2022	Tan et al.	N/A	N/A
2023/0386067	12/2022	De et al.	N/A	N/A
2023/0386153	12/2022	Rybnikov et al.	N/A	N/A
2023/0389991	12/2022	Glaser et al.	N/A	N/A
2023/0394791	12/2022	Wang et al.	N/A	N/A
2023/0397349	12/2022	Capelli et al.	N/A	N/A
2023/0397957	12/2022	Crawford et al.	N/A	N/A
2023/0419496	12/2022	Wuelker et al.	N/A	N/A
2023/0420114	12/2022	Scholler et al.	N/A	N/A
2024/0008935	12/2023	Wolf et al.	N/A	N/A
2024/0016549	12/2023	Johnson et al.	N/A	N/A
2024/0016572	12/2023	Elimelech et al.	N/A	N/A
2024/0020831	12/2023	Johnson et al.	N/A	N/A
2024/0020840	12/2023	Johnson et al.	N/A	N/A
2024/0020862	12/2023	Johnson et al.	N/A	N/A
2024/0022704	12/2023	Benishti et al.	N/A	N/A
2024/0023946	12/2023	Wolf et al.	N/A	N/A
2024/0041530	12/2023	Lang	N/A	N/A
2024/0041558	12/2023	Siewerdsen et al.	N/A	N/A
2024/0045491	12/2023	Sourov	N/A	N/A
2024/0058064	12/2023	Weiser et al.	N/A	N/A
2024/0062387	12/2023	Frantz et al.	N/A	N/A
2024/0103271	12/2023	Zare Seisan	N/A	N/A
2024/0103282	12/2023	Law et al.	N/A	N/A
2024/0111163	12/2023	Law et al.	N/A	N/A
2024/0122560	12/2023	Junio et al.	N/A	N/A
2024/0126087	12/2023	Gera et al.	N/A	N/A
2024/0127559	12/2023	Rybnikov et al.	N/A	N/A
2024/0127578	12/2023	Hiasa	N/A	N/A
2024/0129451	12/2023	Healy et al.	N/A	N/A
2024/0130826	12/2023	Elimelech et al.	N/A	N/A
2024/0134206	12/2023	Gera et al.	N/A	N/A
2024/0144497	12/2023	Cvetko et al.	N/A	N/A
2024/0156532	12/2023	Weiman et al.	N/A	N/A
2024/0177445	12/2023	Galeotti et al.	N/A	N/A
2024/0177458	12/2023	Zhang et al.	N/A	N/A
2024/0180634	12/2023	Mikus	N/A	N/A
2024/0184119	12/2023	Lee et al.	N/A	N/A
2024/0185509	12/2023	Kovler et al.	N/A	N/A
2024/0202926	12/2023	Crawford et al.	N/A	N/A
2024/0202927	12/2023	Haslam et al.	N/A	N/A
2024/0212111	12/2023	Genghi et al.	N/A	N/A

2024/0233131	12/2023	Westerhoff et al.	N/A	N/A
2024/0245463	12/2023	Vilsmeier et al.	N/A	N/A
2024/0245474	12/2023	Weiman et al.	N/A	N/A
2024/0248530	12/2023	Gibby et al.	N/A	N/A
2024/0252252	12/2023	Lang	N/A	N/A
2024/0261036	12/2023	Finley et al.	N/A	N/A
2024/0261058	12/2023	Gera et al.	N/A	N/A
2024/0265645	12/2023	Papar	N/A	N/A
2024/0266033	12/2023	Freeman et al.	N/A	N/A
2024/0268922	12/2023	Calloway et al.	N/A	N/A
2024/0273740	12/2023	Gibby et al.	N/A	N/A
2024/0281979	12/2023	Schrempf et al.	N/A	N/A
2024/0296527	12/2023	Nett et al.	N/A	N/A
2024/0303832	12/2023	Chen et al.	N/A	N/A
2024/0307101	12/2023	Gera et al.	N/A	N/A
2024/0312012	12/2023	Li et al.	N/A	N/A
2024/0341853	12/2023	Gibby et al.	N/A	N/A
2024/0341861	12/2023	Wolf et al.	N/A	N/A
2024/0341910	12/2023	Wolf et al.	N/A	N/A
2024/0341911	12/2023	Elimelech et al.	N/A	N/A
2024/0355098	12/2023	Liu et al.	N/A	N/A
2024/0374314	12/2023	Frey et al.	N/A	N/A
2024/0377640	12/2023	Asaban et al.	N/A	N/A
2024/0378708	12/2023	Kim et al.	N/A	N/A
2024/0382283	12/2023	Kuhnert et al.	N/A	N/A
2024/0386572	12/2023	Barasofsky et al.	N/A	N/A
2024/0386682	12/2023	Cvetko et al.	N/A	N/A
2024/0394883	12/2023	Liao et al.	N/A	N/A
2024/0394985	12/2023	Hanlon et al.	N/A	N/A
2024/0404065	12/2023	Gibbons et al.	N/A	N/A
2024/0404106	12/2023	Wu et al.	N/A	N/A
2024/0404180	12/2023	Kobayashi et al.	N/A	N/A
2024/0420337	12/2023	Li et al.	N/A	N/A
2024/0420592	12/2023	Stone et al.	N/A	N/A
2024/0423724	12/2023	Wolf et al.	N/A	N/A
2024/0423750	12/2023	Elimelech et al.	N/A	N/A
2025/0020931	12/2024	Gera et al.	N/A	N/A
2025/0049534	12/2024	Elimelech et al.	N/A	N/A
2025/0090266	12/2024	Gera et al.	N/A	N/A
2025/0114151	12/2024	Gera et al.	N/A	N/A
2025/0114164	12/2024	Gera et al.	N/A	N/A
2025/0114165	12/2024	Wolf et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3022448	12/2017	CA	N/A
3034314	12/2017	CA	N/A
101379412	12/2008	CN	N/A
102740784	12/2011	CN	N/A

102740789	12/2011	CN	N/A
103106348	12/2012	CN	N/A
103945780	12/2013	CN	N/A
105310756	12/2015	CN	N/A
109199563	12/2018	CN	N/A
111915696	12/2019	CN	N/A
112489047	12/2020	CN	N/A
202004011567	12/2003	DE	N/A
102004011567	12/2004	DE	N/A
102014008153	12/2013	DE	N/A
202022103168	12/2021	DE	N/A
0933096	12/1998	EP	N/A
1640750	12/2005	EP	N/A
1757974	12/2006	EP	N/A
2119397	12/2008	EP	N/A
2134847	12/2008	EP	N/A
2557998	12/2012	EP	N/A
2823463	12/2014	EP	N/A
2868277	12/2014	EP	N/A
2891966	12/2014	EP	N/A
2963616	12/2015	EP	N/A
3028258	12/2015	EP	N/A
3034607	12/2015	EP	N/A
3037038	12/2015	EP	N/A
3069318	12/2015	EP	N/A
3076660	12/2015	EP	N/A
3121789	12/2016	EP	N/A
3123970	12/2016	EP	N/A
2654749	12/2016	EP	N/A
3175815	12/2016	EP	N/A
3216416	12/2016	EP	N/A
2032039	12/2016	EP	N/A
3224376	12/2016	EP	N/A
3247297	12/2016	EP	N/A
3256213	12/2016	EP	N/A
3306567	12/2017	EP	N/A
3320874	12/2017	EP	N/A
2030193	12/2017	EP	N/A
2225723	12/2018	EP	N/A
2619622	12/2018	EP	N/A
2892558	12/2018	EP	N/A
3494903	12/2018	EP	N/A
2635299	12/2018	EP	N/A
3505050	12/2018	EP	N/A
2875149	12/2018	EP	N/A
3593227	12/2019	EP	N/A
3634294	12/2019	EP	N/A
3206583	12/2019	EP	N/A
3711700	12/2019	EP	N/A
2625845	12/2020	EP	N/A

3789965	12/2020	EP	N/A
3858280	12/2020	EP	N/A
3913423	12/2020	EP	N/A
3952331	12/2021	EP	N/A
3960235	12/2021	EP	N/A
3635683	12/2021	EP	N/A
3602492	12/2021	EP	N/A
4173590	12/2022	EP	N/A
3533031	12/2022	EP	N/A
4252695	12/2022	EP	N/A
3195257	12/2022	EP	N/A
3405909	12/2022	EP	N/A
4270313	12/2022	EP	N/A
4287120	12/2022	EP	N/A
3488381	12/2023	EP	N/A
3834768	12/2023	EP	N/A
3903714	12/2023	EP	N/A
4336450	12/2023	EP	N/A
3814984	12/2023	EP	N/A
4115389	12/2023	EP	N/A
3752981	12/2023	EP	N/A
4375948	12/2023	EP	N/A
4383203	12/2023	EP	N/A
4459543	12/2023	EP	N/A
4292045	12/2023	EP	N/A
4298604	12/2023	EP	N/A
2507314	12/2013	GB	N/A
262864	12/2018	IL	N/A
2004-237092	12/2003	JP	N/A
2005-246059	12/2004	JP	N/A
2008-507361	12/2007	JP	N/A
2009-514571	12/2008	JP	N/A
2021-525186	12/2020	JP	N/A
10-2014-0120155	12/2013	KR	N/A
03/34705	12/2002	WO	N/A
2006/002559	12/2005	WO	N/A
2007/051304	12/2006	WO	N/A
2007/115826	12/2006	WO	N/A
2008/103383	12/2007	WO	N/A
2010/067267	12/2009	WO	N/A
2010/074747	12/2009	WO	N/A
2012/061537	12/2011	WO	N/A
2012/101286	12/2011	WO	N/A
2013/112554	12/2012	WO	N/A
2014/014498	12/2013	WO	N/A
2014/024188	12/2013	WO	N/A
2014/037953	12/2013	WO	N/A
2014/113455	12/2013	WO	N/A
2014/125789	12/2013	WO	N/A
2014/167563	12/2013	WO	N/A

2014/174067	12/2013	WO	N/A
2015/058816	12/2014	WO	N/A
2015/061752	12/2014	WO	N/A
2015/109145	12/2014	WO	N/A
2016/151506	12/2015	WO	N/A
2017/042171	12/2016	WO	N/A
2018/052966	12/2017	WO	N/A
2018/073452	12/2017	WO	N/A
2018/200767	12/2017	WO	N/A
2018/206086	12/2017	WO	N/A
2019/083431	12/2018	WO	N/A
2019/135209	12/2018	WO	N/A
2019/135210	12/2018	WO	N/A
2019/161477	12/2018	WO	N/A
2019/195926	12/2018	WO	N/A
2019/210353	12/2018	WO	N/A
2019/211741	12/2018	WO	N/A
2020/109903	12/2019	WO	N/A
2020/109904	12/2019	WO	N/A
2021/017019	12/2020	WO	N/A
2021/019369	12/2020	WO	N/A
2021/021979	12/2020	WO	N/A
2021/023574	12/2020	WO	N/A
2021/046455	12/2020	WO	N/A
2021/048158	12/2020	WO	N/A
2021/061459	12/2020	WO	N/A
2021/062375	12/2020	WO	N/A
2021/073743	12/2020	WO	N/A
2021/087439	12/2020	WO	N/A
2021/091980	12/2020	WO	N/A
2021/112918	12/2020	WO	N/A
2021/130564	12/2020	WO	N/A
2021/137752	12/2020	WO	N/A
2021/141887	12/2020	WO	N/A
2021/145584	12/2020	WO	N/A
2021/154076	12/2020	WO	N/A
2021/183318	12/2020	WO	N/A
2021/188757	12/2020	WO	N/A
2021/255627	12/2020	WO	N/A
2021/257897	12/2020	WO	N/A
2021/258078	12/2020	WO	N/A
2022/009233	12/2021	WO	N/A
2022/053923	12/2021	WO	N/A
2022/056010	12/2021	WO	N/A
2022/079565	12/2021	WO	N/A
2022/180624	12/2021	WO	N/A
2023/003952	12/2022	WO	N/A
2023/281395	12/2022	WO	N/A
2023/007418	12/2022	WO	N/A
2023/011924	12/2022	WO	N/A

2023/021448	12/2022	WO	N/A
2023/021450	12/2022	WO	N/A
2023/021451	12/2022	WO	N/A
2023/026229	12/2022	WO	N/A
2023/047355	12/2022	WO	N/A
2023/072887	12/2022	WO	N/A
2023/088986	12/2022	WO	N/A
2023/158878	12/2022	WO	N/A
2023/159104	12/2022	WO	N/A
2023/161848	12/2022	WO	N/A
2023/163933	12/2022	WO	N/A
2023/175244	12/2022	WO	N/A
2023/186996	12/2022	WO	N/A
2023/202909	12/2022	WO	N/A
2023/205212	12/2022	WO	N/A
2023/205896	12/2022	WO	N/A
2023/209014	12/2022	WO	N/A
2023/229415	12/2022	WO	N/A
2023/232492	12/2022	WO	N/A
2023/240912	12/2022	WO	N/A
2024/001140	12/2023	WO	N/A
2024/002620	12/2023	WO	N/A
2024/013642	12/2023	WO	N/A
2024/018368	12/2023	WO	N/A
2024/046760	12/2023	WO	N/A
2024/052136	12/2023	WO	N/A
2024/077077	12/2023	WO	N/A
2024/121060	12/2023	WO	N/A
2024/132609	12/2023	WO	N/A
2024/145341	12/2023	WO	N/A
2024/160896	12/2023	WO	N/A
2024/165508	12/2023	WO	N/A
2024/173251	12/2023	WO	N/A
2024/186811	12/2023	WO	N/A
2024/226797	12/2023	WO	N/A
2024/251344	12/2023	WO	N/A

OTHER PUBLICATIONS

Lee, Sing Chun, et al. "Multi-modal imaging, model-based tracking, and mixed reality visualisation for orthopaedic surgery." Healthcare technology letters 4.5 (2017): 168-173. cited by examiner

International Search Report and Written Opinion received for PCT Patent Application No. PCT/IB2022/057733, dated Jan. 26, 2023, 14 pages. cited by applicant

U.S. Appl. No. 16/419,023 (U.S. Pat. No. 11,750,794, filed May 22, 2019, Combining Video-Based and Optic-Based Augmented Reality in a Near Eye Display. cited by applicant

U.S. Appl. No. 18/365,643, filed Aug. 4, 2023, Head-Mounted Augmented Reality Near Eye Display Device. cited by applicant

U.S. Appl. No. 18/365,650, filed Aug. 4, 2023, Systems for Facilitating Augmented Reality-Assisted Medical Procedures. cited by applicant

U.S. Appl. No. 18/365,590, filed Aug. 4, 2023, Registration of a Fiducial Marker for an Augmented Reality System. cited by applicant

U.S. Appl. No. 18/365,571, filed Aug. 4, 2023, Registration Marker for an Augmented Reality System. cited by applicant

U.S. Appl. No. 17/045,766, filed Oct. 7, 2020, Registration of a Fiducial Marker for an Augmented Reality System. cited by applicant

U.S. Appl. No. 16/524,258, filed Jul. 29, 2019, Fiducial Marker. cited by applicant

U.S. Appl. No. 17/827,710, filed May 29, 2022, Mirroring in Image Guided Surgery. cited by applicant

U.S. Appl. No. 16/200,144, filed Nov. 26, 2018, Tracking System for Image-Guided Surgery. cited by applicant

U.S. Appl. No. 18/044,380, filed Mar. 8, 2023, Universal Tool Adapter for Image-Guided Surgery. cited by applicant

U.S. Appl. No. 17/368,859, filed Jul. 7, 2021, Iliac Pin and Adapter. cited by applicant

U.S. Appl. No. 35/508,942 (U.S. Pat. No. D. 930,162, filed Feb. 13, 2020 (Sep. 7, 2021), Medical Headset. cited by applicant

16 Augmented Reality Glasses of 2021 (with Features), in Back to News, Dated May 6, 2022, accessed at <https://web.archive.org/web/20221127195438/https://circuitstream.com/blog/16-augmented-reality-glasses-of-2021-with-features-breakdowns/>. cited by applicant

EverySight, Installing your RX Adaptor, accessed Mar. 13, 2024 at <https://support.eversight.com/hc/en-us/articles/115000984571-Installing-your-RX-Adaptor>. cited by applicant

EverySight, Raptor User Manual, copyright 2017, in 46 pages. cited by applicant

Frames Direct, InSpatialRx Prescription Insert, Prescription Insert for Magic Leap 1, accessed Mar. 8, 2024 at <https://www.framesdirect.com/inspatialrx-prescription-insert.html>. cited by applicant

Reddit, Notice on Prescription Lenses for Nreal Glasses, accessed Mar. 13, 2024 at https://www.reddit.com/r/nreal/comments/x1fte5/notice_on_prescription_lenses_for_nreal_glasses/. cited by applicant

Vuzix Blades, Prescription Lens Installation Guide, copyright 2020. cited by applicant

U.S. Appl. No. 15/896,102, U.S. Pat. No. 10,134,166, filed Feb 14, 2018, Nov. 20, 2018, Combining Video-Based and Optic-Based Augmented Reality In A Near Eye Display. cited by applicant

U.S. Appl. No. 16/159,740, U.S. Pat. No. 10,382,748, filed Oct. 15, 2018, Aug. 13, 2019, Combining Video-Based and Optic-Based Augmented Reality in a Near Eye Display. cited by applicant

U.S. Appl. No. 16/419,023, U.S. Pat. No. 11,750,794, filed May 22, 2019, Sep. 5, 2023, Combining Video-Based and Optic-Based Augmented Reality in a Near Eye Display. cited by applicant

U.S. Appl. No. 18/352,158, filed Jul. 13, 2023, Combining Video-Based and Optic-Based Augmented Reality in a Near Eye Display. cited by applicant

U.S. Appl. No. 18/365,643, U.S. Pat. No. 12,069,233, filed Aug. 4, 2023, Head-Mounted Augmented Reality Near Eye Display Device. cited by applicant

U.S. Appl. No. 18/365,650, U.S. Pat. No. 12,063,345, filed Aug. 4, 2023, Systems For Facilitating Augmented Reality-Assisted Medical Procedures. cited by applicant

U.S. Appl. No. 15/127,423, U.S. Pat. No. 9,928,629, filed Sep. 20, 2016, Mar. 27, 2018, Combining Video-Based and Optic-Based Augmented Reality in a Near Eye Display. cited by applicant

U.S. Appl. No. 16/120,480, U.S. Pat. No. 10,835,296, filed Sep. 4, 2018, Nov. 17, 2020, Spinous Process Clamp. cited by applicant

U.S. Appl. No. 17/067,831, filed Oct. 12, 2020, Spinous Process Clamp. cited by applicant

U.S. Appl. No. 18/030,072, filed Apr. 4, 2023, Spinous Process Clamp. cited by applicant

U.S. Appl. No. 18/365,590, U.S. Pat. No. 11,980,508, filed Aug. 4, 2023, May 14, 2024, Registration of a Fiducial Marker for an Augmented Reality System. cited by applicant

U.S. Appl. No. 18/365,571, U.S. Pat. No. 11,974,887, filed Aug. 4, 2023, May 7, 2024, Registration

Marker for an Augmented Reality System. cited by applicant
U.S. Appl. No. 18/632,588, filed Apr. 11, 2024, Registration of a Fiducial Marker for an Augmented Reality System. cited by applicant
U.S. Appl. No. 17/045,766, U.S. Pat. No. 11,980,507, filed Oct. 7, 2020, May 14, 2024, Registration of a Fiducial Marker for an Augmented Reality System. cited by applicant
U.S. Appl. No. 16/199,281, U.S. Pat. No. 10,939,977, filed Nov. 26, 2018, Mar. 9, 2021, Positioning Marker. cited by applicant
U.S. Appl. No. 16/524,258, U.S. Pat. No. 11,980,506, filed Jul. 29, 2019, May 14, 2024, Fiducial Marker. cited by applicant
U.S. Appl. No. 18/631,804, filed Apr. 10, 2024, Fiducial Marker. cited by applicant
U.S. Appl. No. 17/585,629, filed Jan. 27, 2022, Fiducial Marker. cited by applicant
U.S. Appl. No. 16/724,297, U.S. Pat. No. 11,382,712, filed Dec. 22, 2019, Jul. 12, 2022, Mirroring in Image Guided Surgery. cited by applicant
U.S. Appl. No. 17/827,710, U.S. Pat. No. 11,801,115, filed May 29, 2022, Oct. 31, 2023, Mirroring in Image Guided Surgery. cited by applicant
U.S. Appl. No. 18/352,181, filed Jul. 13, 2023, Mirroring in Image Guided Surgery. cited by applicant
U.S. Appl. No. 18/400,739, filed Dec. 29, 2023, Mirroring in Image Guided Surgery. cited by applicant
U.S. Appl. No. 18/780,095, filed Jul. 22, 2024, Mirroring in Image Guided Surgery. cited by applicant
U.S. Appl. No. 16/200,144, U.S. Pat. No. 11,766,296, filed Nov. 26, 2018, Sep. 26, 2023, Tracking System for Image-Guided Surgery. cited by applicant
U.S. Appl. No. 18/470,809, U.S. Pat. No. 11,980,429, filed Sep. 20, 2023, May 14, 2024, Tracking Methods for Image-Guided Surgery. cited by applicant
U.S. Appl. No. 18/631,877, filed Apr. 10, 2024, Tracking Systems and Methods for Image-Guided Surgery. cited by applicant
U.S. Appl. No. 17/015,199, filed Sep. 9, 2020, Universal Tool Adapter. cited by applicant
U.S. Appl. No. 18/598,965, filed Mar. 7, 2024, Universal Tool Adapter For Image Guide Surgery. cited by applicant
U.S. Appl. No. 18/044,380, filed Mar. 8, 2023, Universal Tool Adapter For Image-Guide Surgery. cited by applicant
U.S. Appl. No. 16/901,026, U.S. Pat. No. 11,389,252, filed Jun. 15, 2020, Jul. 19, 2022, Rotating Marker For Image Guided Surgery. cited by applicant
U.S. Appl. No. 18/008,980, filed Dec. 8, 2022, Rotating Marker. cited by applicant
U.S. Appl. No. 17/368,859, U.S. Pat. No. 11,896,445, filed Jul. 7, 2021, Feb. 13, 2024, Iliac Pin and Adapter. cited by applicant
U.S. Appl. No. 18/437,898, filed Feb. 9, 2024, Iliac Pin and Adapter. cited by applicant
U.S. Appl. No. 18/576,516, filed Jan. 4, 2024, Iliac Pin and Adapter. cited by applicant
U.S. Appl. No. 17/388,064, filed Jul. 29, 2021, Rotating Marker and Adapter for Image-Guided Surgery. cited by applicant
U.S. Appl. No. 18/291,731, filed Jan. 24, 2024, Rotating Marker and Adapter for Image-Guided Surgery. cited by applicant
U.S. Appl. No. 18/365,844, filed Aug. 4, 2023, Augmented-Reality Surgical System Using Depth Sensing. cited by applicant
U.S. Appl. No. 18/683,676, filed Feb. 14, 2024, Stereoscopic Display and Digital Loupe for Augmented-Reality Near-Eye Display. cited by applicant
U.S. Appl. No. 18/683,680, filed Feb. 14, 2024, Augmented Reality Assistance for Osteotomy and Discectomy. cited by applicant
U.S. Appl. No. 18/684,756, filed Feb. 19, 2024, Registration and Registration Validation in Image-

Guided Surgery. cited by applicant

U.S. Appl. No. 18/693,338, filed Mar. 19, 2024, Surgical Planning and Display. cited by applicant

U.S. Appl. No. 18/365,566, filed Aug. 4, 2023, Systems for Medical Image Visualization. cited by applicant

U.S. Appl. No. 18/399,253, filed Dec. 28, 2023, Methods for Medical Image Visualization. cited by applicant

U.S. Appl. No. 18/398,837, U.S. Pat. No. 12,044,858, filed Dec. 28, 2023, Jul. 23, 2024, Adjustable Augmented Reality Eyewear for Image-Guided Medical Intervention. cited by applicant

U.S. Appl. No. 18/399,433, U.S. Pat. No. 12,044,856, filed Dec. 28, 2023, Jul. 23, 2024, Configurable Augmented Reality Eyewear for Image-Guided Medical Intervention. cited by applicant

U.S. Appl. No. 18/772,578, filed Jul. 15, 2024, Augmented Reality Eyewear for Image-Guided Medical Intervention. cited by applicant

U.S. Appl. No. 35/508,942, U.S. Pat. No. 930,162, filed Feb. 13, 2020, Sep. 7, 2021, Medical Headset. cited by applicant

Augmedics Ltd., 510k Clearance Summary for Augmedics' xvision Spine system, dated Dec. 20, 2019 in 11 pages. cited by applicant

Medtronic Navigation, Inc., StealthStation™ S8 System Manual in 82 pages, Revision 2, Copyright 2018. cited by applicant

Novarad Healthcare IT and Imaging, OpenSight English: See 3D Medical Images Using Augmented Reality, dated Mar. 9, 2018, accessed via YouTube on Mar. 11, 2025 at https://www.youtube.com/watch?v=M3yY_b8jT54. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International PCT Application PCT/IB2022/057733, filed Aug. 18, 2022, which claims the benefit of: U.S. Provisional Patent Application 63/236,241, filed Aug. 24, 2021; U.S. Provisional Patent Application 63/281,677, filed Nov. 21, 2021; U.S. Provisional Patent Application No. 63/234,272, filed Aug. 18, 2021; and U.S. Provisional Patent Application No. 63/236,244, filed Aug. 24, 2021. The entire content of each of these related applications is incorporated herein by reference.

FIELD

(1) The present disclosure relates generally to image-guided surgery or intervention, and specifically to systems and methods for use of augmented reality in image-guided surgery or intervention and/or to systems and methods for use in surgical computer-assisted navigation.

BACKGROUND

(2) Near-eye display devices and systems can be used in augmented reality systems, for example, for performing image-guided surgery. In this way, a computer-generated image may be presented to a healthcare professional who is performing the procedure such that the image is aligned with an anatomical portion of a patient who is undergoing the procedure. Applicant's own work has demonstrated an image of a tool that is used to perform the procedure can also be incorporated into the image that is presented on the head-mounted display. For example, Applicant's prior systems for image-guided surgery have been effective in tracking the positions of the patient's body and the tool (see, for example, U.S. Pat. Nos. 9,928,629, 10,835,296, 10,939,977, PCT International

SUMMARY

(3) Embodiments of the present disclosure provide improved systems, methods, and software for image-guided surgery. Some embodiments of the systems improve the accuracy of the augmented-reality images that are presented on the display and broaden the capabilities of the augmented reality system by employing depth sensing.

(4) In some embodiments, a system for image-guided surgery comprises a head-mounted unit, comprising a see-through augmented-reality display and a depth sensor, which is configured to generate depth data with respect to a region of interest (ROI) of a body of a patient that is viewed through the display by a user wearing the head-mounted unit; and a processor, which is configured to receive a three-dimensional (3D) tomographic image of the body of the patient, to compute a depth map of the ROI based on the depth data generated by the depth sensor, to compute a transformation over the ROI so as to register the tomographic image with the depth map, and to apply the transformation in presenting a part of the tomographic image on the display in registration with the ROI viewed through the display.

(5) In some embodiments, the ROI comprises a bone of the body to which an anchoring device is fastened, and wherein the processor is further configured to identify a location of the anchoring device in the depth map, to update the depth map in the course of a surgery, to detect a change in the location of the anchoring device in the updated depth map, and to take a corrective action responsively to the change.

(6) In some embodiments, the corrective action comprises modifying a presentation on the display responsively to the change in the location of the anchoring device.

(7) In some embodiments, the depth map includes a spine of the patient, which is exposed in a surgical procedure, and wherein the processor is configured to compute the transformation by registering the spine in the depth map with the spine appearing in the tomographic image.

(8) In some embodiments, the processor is configured to process the depth data so as to detect a position of a marker that is fixed to the body of the patient, to recognize a location of the head-mounted unit relative to the body based on the detected position, and to position the image presented on the display responsively to the recognized location.

(9) In some embodiments, the processor is configured to process the depth data so as to identify a change in an anatomical structure in the body of the patient during a surgical procedure, and to modify the image presented on the display responsively to the identified change.

(10) In some embodiments, the processor is configured to process the depth data so as to identify an implant inserted into the body of the patient during a surgical procedure, and to modify the image presented on the display responsively to the identified implant.

(11) In some embodiments, the tomographic image comprises a CT scan of the patient, which was performed with an array of radiopaque fiducial markers fixed to the body of the patient, and wherein the processor is configured to identify respective 3D coordinates of the fiducial markers in the depth map and to register the CT scan with the ROI viewed through the display by matching the fiducial markers in the CT to the respective 3D coordinates.

(12) In some embodiments, a system for image-guided surgery comprises a head-mounted unit, comprising: a see-through augmented-reality display; and a depth sensor, which is configured to generate depth data with respect to a region of interest (ROI) on a body of a patient that is viewed through the display by a user wearing the head-mounted unit and with respect to a surgical tool when the tool is placed within a field of view of the depth sensor, wherein the tool comprises a shaft and a marker containing a predefined pattern disposed on the tool in a fixed spatial relation to the shaft; and a processor, which is configured to: process the depth data so as to identify a shape of the tool and to compute, responsively to the shape, a spatial transformation between a position of the marker and a location and orientation of the shaft; track the position of the marker as the user

manipulates the shaft of the tool within the body, and using the tracked position and the spatial transformation, generate an image of the tool, including the shaft, on the display in registration with the ROI viewed through the display.

(13) In some embodiments, further comprising a tracking sensor, which is disposed on the head-mounted unit in a known spatial relation to the depth sensor and is configured to detect the position of the marker.

(14) In some embodiments, the shaft has a curved shape, and wherein the processor is configured to process the depth data so as to reconstruct a three-dimensional (3D) model of the curved shape, and to generate the image of the tool based on the 3D model.

(15) In some embodiments, the processor is configured to process the depth data so as to detect a change in a shape of the tool and to update the image of the tool on the display responsively to the change in the shape.

(16) In some embodiments, the depth sensor is further configured to generate the depth data with respect to a further marker that is attached to the body of the patient, and wherein the processor is configured to apply the depth data in calculating a position of the tool relative to the body.

(17) In some embodiments, the depth sensor is configured to generate further depth data with respect to a hand of a user of the head-mounted unit, and wherein the processor is configured to process the further depth data so as to detect a gesture made by the hand, and to control a function of the system responsively to the detected gesture.

(18) In some embodiments, a system for image-guided surgery, comprises a head-mounted unit, comprising a see-through augmented-reality display and a depth sensor, which is configured to generate depth data with respect to a region of interest (ROI) on a body of a patient that is viewed through the display by a user wearing the head-mounted unit and with respect to a surgical implant when the implant is placed within a field of view of the depth sensor, wherein the implant is configured to be mounted on a shaft of a surgical tool and inserted, using the tool, into the body; and a processor, which is configured to process the depth data so as to identify a shape of the implant and to compute, responsively to the shape, a spatial transformation between a position of a marker disposed on the tool and a location and orientation of the implant, to track the position of the marker as the user manipulates the shaft of the tool within the body, and using the tracked position, the spatial transformation, and the identified shape, to generate on the display an image of the implant within the body in registration with the ROI viewed through the display, wherein the marker contains a predefined pattern and is disposed in a fixed spatial relation to the shaft.

(19) In some embodiments, the system further comprising a tracking sensor, which is disposed on the head-mounted unit in a known spatial relation to the depth sensor and is configured to detect the position of the marker.

(20) In some embodiments, the shaft has a curved shape, and wherein the processor is configured to process the depth data so as to reconstruct a three-dimensional (3D) model of the curved shape, and to generate the image of the implant based on the 3D model.

(21) In some embodiments, the processor is configured to process the depth data so as to detect a change in a shape of the tool and to update the spatial transformation responsively to the change in the shape.

(22) In some embodiments, a system for image-guided surgery, comprises a head-mounted unit, comprising a see-through augmented-reality display and a depth sensor, which is configured to generate depth data with respect to a region of interest (ROI) on a body of a patient, including a bone inside the body, that is viewed through the display by a user wearing the head-mounted unit; and a processor, which is configured to process the depth data generated by the depth sensor so as to identify a first three-dimensional (3D) shape of the bone prior to a surgical procedure on the bone and a second 3D shape of the bone following the surgical procedure, and to generate, based on the first and second 3D shapes, an image showing a part of the bone that was removed in the surgical procedure.

- (23) In some embodiments, a method for image-guided surgery comprises using a head-mounted unit that includes a see-through augmented-reality display and a depth sensor, generating depth data with respect to a region of interest (ROI) of a body of a patient that is viewed through the display by a user wearing the head-mounted unit; receiving a three-dimensional (3D) tomographic image of the body of the patient; computing a depth map of the ROI based on the depth data generated by the depth sensor; computing a transformation over the ROI so as to register the tomographic image with the depth map; and applying the transformation in presenting a part of the tomographic image on the display in registration with the ROI viewed through the display.
- (24) In some embodiments, the ROI comprises a bone of the body to which an anchoring device is fastened, and wherein the method comprises: identifying an initial location of the anchoring device in the depth map; updating the depth map in the course of a surgery; detecting a change in the location of the anchoring device in the updated depth map; and taking a corrective action responsively to the change.
- (25) In some embodiments, taking the corrective action comprises modifying a presentation on the display responsively to the change in the location of the anchoring device.
- (26) In some embodiments, the depth map includes a spine of the patient, which is exposed in a surgical procedure, and wherein computing the transformation comprises registering the spine in the depth map with the spine appearing in the tomographic image.
- (27) In some embodiments, the method further comprises processing the depth data so as to detect a position of a marker that is fixed to the body of the patient; recognizing a location of the head-mounted unit relative to the body based on the detected position; and positioning the image presented on the display responsively to the recognized location.
- (28) In some embodiments, the method further comprises processing the depth data so as to identify a change in an anatomical structure in the body of the patient during a surgical procedure; and modifying the image presented on the display responsively to the identified change.
- (29) In some embodiments, the method further comprises processing the depth data so as to identify an implant inserted into the body of the patient during a surgical procedure; and modifying the image presented on the display responsively to the identified implant.
- (30) In some embodiments, the tomographic image comprises a CT scan of the patient, which was performed with an array of radiopaque fiducial markers fixed to the body of the patient, and wherein computing the transformation comprises identifying respective 3D coordinates of the fiducial markers in the depth map, and registering the CT scan with the ROI viewed through the display by matching the fiducial markers in the CT to the respective 3D coordinates.
- (31) In some embodiments, a method for image-guided surgery comprises using a head-mounted unit that includes a see-through augmented-reality display and a depth sensor, generating depth data with respect to a region of interest (ROI) on a body of a patient that is viewed through the display by a user wearing the head-mounted unit and with respect to a surgical tool when the tool is placed within a field of view of the depth sensor, wherein the tool comprises a shaft and a marker containing a predefined pattern disposed on the tool in a fixed spatial relation to the shaft; processing the depth data so as to identify a shape of the tool and to compute, responsively to the shape, a spatial transformation between a position of the marker and a location and orientation of the shaft; tracking the position of the marker as the user manipulates the shaft of the tool within the body; and using the tracked position and the spatial transformation, generating an image of the tool, including the shaft, on the display in registration with the ROI viewed through the display.
- (32) In some embodiments, the tracking the position comprises detecting the position of the marker using a tracking sensor disposed on the head-mounted unit in a known spatial relation to the depth sensor.
- (33) In some embodiments, the shaft has a curved shape, and wherein processing the depth data comprises reconstructing a three-dimensional (3D) model of the curved shape, wherein the image of the tool is generated based on the 3D model.

- (34) In some embodiments, processing the depth data comprises detecting a change in a shape of the tool, and wherein generating the image comprises updating the image of the tool on the display responsively to the change in the shape.
- (35) In some embodiments, generating the depth data comprises capturing further depth data with respect to a further marker that is attached to the body of the patient, and wherein processing the depth data comprises applying the further depth data in calculating a position of the tool relative to the body.
- (36) In some embodiments, generating the depth data comprises capturing further depth data with respect to a hand of a user of the head-mounted unit, and wherein the method comprises processing the further depth data so as to detect a gesture made by the hand, and controlling a function of the head-mounted unit responsively to the detected gesture.
- (37) In some embodiments, a method for image-guided surgery comprises using a head-mounted unit that includes a see-through augmented-reality display and a depth sensor, generating depth data with respect to a region of interest (ROI) on a body of a patient that is viewed through the display by a user wearing the head-mounted unit and with respect to a surgical implant when the implant is placed within a field of view of the depth sensor, wherein the implant is mounted on a shaft of a surgical tool and inserted, using the tool, into the body; processing the depth data so as to identify a shape of the implant; computing, responsively to the shape, a spatial transformation between a position of a marker disposed on the tool and a location and orientation of the implant, wherein the marker contains a predefined pattern and is disposed in a fixed spatial relation to the shaft; tracking the position of the marker as the user manipulates the shaft of the tool within the body; and using the tracked position, the spatial transformation, and the identified shape, generating on the display an image of the implant within the body in registration with the ROI viewed through the display.
- (38) In some embodiments, the method further comprises detecting the position of the marker using a tracking sensor, which is disposed on the head-mounted unit in a known spatial relation to the depth sensor.
- (39) In some embodiments, the shaft has a curved shape, and wherein processing the depth data comprises reconstructing a three-dimensional (3D) model of the curved shape, wherein the image of the implant is generated based on the 3D model.
- (40) In some embodiments, processing the depth data comprises detecting a change in a shape of the tool, and updating the spatial transformation responsively to the change in the shape.
- (41) In some embodiments, a method for image-guided surgery, comprises using a head-mounted unit that includes a see-through augmented-reality display and a depth sensor, generating depth data with respect to a region of interest (ROI) on a body of a patient, including a bone inside the body, that is viewed through the display by a user wearing the head-mounted unit; processing the depth data generated by the depth sensor so as to identify a first three-dimensional (3D) shape of the bone prior to a surgical procedure on the bone and a second 3D shape of the bone following the surgical procedure; and generating, based on the first and second 3D shapes, an image showing a part of the bone that was removed in the surgical procedure.
- (42) In some embodiments, the surgical procedure involves a bone cut.
- (43) In some embodiments, a head-mounted system for image-guided surgery comprises a see-through augmented-reality display disposed so as to be viewable by a user over a region of interest (ROI) of a body of a patient; a depth sensor configured to generate depth data with respect to the ROI; and a processor and a memory for storing instructions that, when executed by the processor cause the system to: receive a three-dimensional (3D) tomographic image of the body of the patient; determine a depth map of the ROI based at least in part on the depth data; determine a transformation over the ROI so as to register the 3D tomographic image with the depth map; and display at least a part of the 3D tomographic image on the see-through augmented-reality display in registration with the ROI based at least in part on the transformation.

(44) In some embodiments, a method for image-guided surgery comprises a see-through augmented-reality display disposed so as to be viewable by a user over a region of interest (ROI) of a body of a patient; a depth sensor configured to generate depth data with respect to the ROI; and a processor and a memory for storing instructions that, when executed by the processor cause the system to: receive a three-dimensional (3D) tomographic image of the body of the patient; determine a depth map of the ROI based at least in part on the depth data; determine a transformation over the ROI so as to register the 3D tomographic image with the depth map; and display at least a part of the 3D tomographic image on the see-through augmented-reality display in registration with the ROI based at least in part on the transformation.

(45) For purposes of summarizing the disclosure, certain aspects, advantages, and novel features are discussed herein. It is to be understood that not necessarily all such aspects, advantages, or features will be embodied in any particular embodiment of the disclosure, and an artisan would recognize from the disclosure herein a myriad of combinations of such aspects, advantages, or features.

(46) The present invention will be more fully understood from the following detailed description of the embodiments thereof, taken together with the drawings in which:

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Non-limiting features of some embodiments of the invention are set forth with particularity in the claims that follow. The following drawings are for illustrative purposes only and show non-limiting embodiments. Features from different figures may be combined in several embodiments.

(2) FIG. 1 is a schematic pictorial illustration showing a system for image-guided surgery, in accordance with an embodiment of the disclosure;

(3) FIG. 2A is a schematic pictorial illustration showing details of a near-eye unit that is used for image-guided surgery, in accordance with an embodiment of the disclosure;

(4) FIG. 2B is a schematic pictorial illustration showing details of a head-mounted unit that is used for image-guided surgery, in accordance with another embodiment of the disclosure;

(5) FIG. 3A is a schematic pictorial illustration showing details of a surgical tool, in accordance with an embodiment of the disclosure;

(6) FIG. 3B is a schematic pictorial illustration showing details of a surgical tool, in accordance with another embodiment of the disclosure;

(7) FIGS. 4A and 4B are flow charts that schematically illustrate methods for image-guided surgery, in accordance with embodiments of the disclosure;

(8) FIGS. 5, 6 and 7 are flow charts that schematically illustrate methods for image-guided surgery, in accordance with further embodiments of the disclosure; and

(9) FIG. 8 is a schematic pictorial illustration showing a system for image-guided surgery, in accordance with an alternative embodiment of the disclosure.

DETAILED DESCRIPTION

Overview

(10) Embodiments of the present disclosure that are described herein provide systems, methods and software for image-guided surgery or other intervention, computer assisted navigation and/or stereotactic surgery or other intervention that, inter alia, use depth sensing to enhance the capabilities of an augmented-reality display and system. In some embodiments, a head-mounted unit comprises both a see-through augmented-reality display and a depth sensor. In some embodiments, the depth sensor generates depth data with respect to a region of interest (ROI) on a body of a patient that is viewed through the display by a user wearing the head-mounted unit. In some embodiments, the system applies the depth data in generating one or more depth maps of the

body. Additionally or alternatively, the depth sensor may be applied in generating depth data with respect to implements such as clamps, tools and implants that can be inserted into the body. In some embodiments, using the depth data, the system is able to improve the accuracy of the augmented-reality images that are presented on the display and broaden the capabilities of the augmented-reality system.

(11) In some embodiments, the term “depth sensor” refers to one or more optical components that are configured to capture a depth map of a scene. For example, in some embodiments, the depth sensor can be a pattern projector and a camera for purposes of structured-light depth mapping. For example, in some embodiments, the depth sensor can be a pair of cameras configured for stereoscopic depth mapping. For example, in some embodiments, the depth sensor can be a beam projector and a detector (or an array of detectors) configured for time-of-flight measurement. Of course the term “depth sensor” as used herein is not limited to the listed examples and can other structure.

System Description

(12) Reference is now made to FIGS. **1** and **2A**, which schematically illustrate an exemplary system **10** for image-guided surgery, in accordance with some embodiments of the disclosure. For example, FIG. **1** is a pictorial illustration of the system **10** as a whole, while FIG. **2A** is a pictorial illustration of a near-eye unit that is used in the system **10**. The near eye unit illustrated in FIGS. **1** and **2A** is configured as a head-mounted unit **28**. In some embodiments, the near-eye unit can be configured as the head-mounted unit **28** shown in FIG. **8** and as a head-mounted AR display (HMD) unit **70**, **100** and described hereinbelow. In FIG. **1**, the system **10** is applied in a medical procedure on a patient **20** using image-guided surgery. In this procedure, a tool **22** is inserted via an incision in the patient's back in order to perform a surgical intervention. Alternatively, the system **10** and the techniques described herein may be used, mutatis mutandis, in other surgical procedures.

(13) Methods for optical depth mapping can generate a three-dimensional (3D) profile of the surface of a scene by processing optical radiation reflected from the scene. In the context of the present description and in the claims, the terms depth map, 3D profile, and 3D image are used interchangeably to refer to an electronic image in which the pixels contain values of depth or distance from a reference point, instead of or in addition to values of optical intensity.

(14) In some embodiments, depth mapping systems can use structured light techniques in which a known pattern of illumination is projected onto the scene. Depth can be calculated based on the deformation of the pattern in an image of the scene. In some embodiments, depth mapping systems use stereoscopic techniques, in which the parallax shift between two images captured at different locations is used to measure depth. In some embodiments, depth mapping systems can sense the times of flight of photons to and from points in the scene in order to measure the depth coordinates. In some embodiments, depth mapping systems control illumination and/or focus and can use various sorts of image processing techniques.

(15) In the embodiment illustrated in FIG. **1**, a user of the system **10**, such as a healthcare professional **26** (for example, a surgeon performing the procedure), wears the head-mounted unit **28**. In some embodiments, the head-mounted unit **28** includes one or more see-through displays **30**, for example as described in the above-mentioned U.S. Pat. No. 9,928,629 or in the other patents and applications cited above.

(16) In some embodiments, the one or more see-through displays **30** include an optical combiner. In some embodiments, the optical combiner is controlled by one or more processors **32**. In some embodiments, the one or more processors **32** is disposed in a central processing system **50**. In some embodiments, the one or more processors **32** is disposed in the head-mounted unit **28**. In some embodiments, the one or more processors **32** are disposed in both the central processing system **50** and the head-mounted unit **28** and can share processing tasks and/or allocate processing tasks between the one or more processors **32**.

(17) In some embodiments, the one or more see-through displays **30** display an augmented-reality image to the healthcare professional **26**. In some embodiments, the augmented reality image viewable through the one or more see-through displays **30** is a combination of objects visible in the real world with the computer-generated image. In some embodiments, each of the one or more see-through displays **30** comprises a first portion **33** and a second portion **35**. In some embodiments, the one or more see-through displays **30** display the augmented-reality image such that the computer-generated image is projected onto the first portion **33** in alignment with the anatomy of the body of the patient **20** that is visible to the healthcare professional **26** through the second portion **35**.

(18) In some embodiments, the computer-generated image includes a virtual image of one or more tools **22**. In some embodiments, the system **10** combines at least a portion of the virtual image of the one or more tools **22** into the computer-generated image. For example, some or all of the tool **22** may not be visible to the healthcare professional **26** because, for example, a portion of the tool **22** is hidden by the patient's anatomy (e.g., a distal end of the tool **22**). In some embodiments, the system **10** can display the virtual image of at least the hidden portion of the tool **22** as part of the computer-generated image displayed in the first portion **33**. In this way, the virtual image of the hidden portion of the tool **22** is displayed on the patient's anatomy. In some embodiments, the portion of the tool **22** hidden by the patient's anatomy increase and/or decreases over time or during the procedure. In some embodiments, the system **10** increase and/or decreases the portion of the tool **22** included in the computer-generated image based on the changes in the portion of the tool **22** hidden by the patient's anatomy over time.

(19) Some embodiments of the system **10** comprise an anchoring device (e.g., bone marker **60**) for indicating the body of the patient **20**. For example, in image-guided surgery and other surgeries that utilize the system **10**, the bone marker **60** can be used as a fiducial marker. In some embodiments, the bone marker **60** can be coupled with the fiducial marker. In system **10**, for example, the anchoring device is configured as the bone marker **60** (e.g., anchoring device coupled with a marker that is used to register an ROI of the body of the patient **20**). In some embodiments, the anchoring device is coupled with a tracking system, with a preoperative or intraoperative CT scan of the ROI. During the procedure, in some embodiments, the tracking system (for example an IR tracking system) tracks the marker mounted on the anchoring device and the tool **22** mounted with a tool marker **40**. Following that, the display of the CT image data, including, for example, a model generated based on such data, on the near-eye display may be aligned with the surgeon's actual view of the ROI based on this registration. In addition, a virtual image of the tool **22** may be displayed on the CT model based on the tracking data and the registration. The user may then navigate the tool **22** based on the virtual display of the tool **22** with respect to the patient image data, and optionally, while it is aligned with the user view of the patient or ROI.

(20) According to some aspects, the image presented on the one or more see-through displays **30** is aligned with the body of the patient **20**. According to some aspects, misalignment of the image presented on the one or more see-through displays **30** with the body of the patient **20** may be allowed. In some embodiments, the misalignment may be 0-1 mm, 1-2 mm, 2-3 mm, 3-4 mm, 4-5 mm, 5-6 mm, and overlapping ranges therein. According to some aspects, the misalignment may typically not be more than about 5 mm. In order to account for such a limit on the misalignment of the patient's anatomy with the presented images, the position of the patient's body, or a portion thereof, with respect to the head-mounted unit **28** can be tracked. For example, in some embodiments, a patient marker **38** and/or the bone marker **60** attached to an anchoring implement or device such as a clamp **58** or pin, for example, may be used for this purpose, as described further hereinbelow.

(21) When an image of the tool **22** is incorporated into the computer-generated image that is displayed on the head-mounted unit **28** or the HMD unit **70**, the position of the tool **22** with respect to the patient's anatomy should be accurately reflected. For this purpose, the position of the tool **22**

or a portion thereof, such as the tool marker **40**, is tracked by the system **10**. In some embodiments, the system **10** determines the location of the tool **22** with respect to the patient's body such that errors in the determined location of the tool **22** with respect to the patient's body are reduced. For example, in certain embodiments, the errors may be 0-1 mm, 1-2 mm, 2-3 mm, 3-4 mm, 4-5 mm, and overlapping ranges therein.

(22) In some embodiments, the head-mounted unit **28** includes a tracking sensor **34** to facilitate determination of the location and orientation of the head-mounted unit **28** with respect to the patient's body and/or with respect to the tool **22**. In some embodiments, tracking sensor **34** can also be used in finding the position and orientation of the tool **22** and the clamp **58** with respect to the patient's body. In some embodiments, the tracking sensor **34** comprises an image-capturing device **36**, such as a camera, which captures images of the patient marker **38**, the bone marker **60**, and/or the tool marker **40**. For some applications, an inertial-measurement unit **44** is also disposed on the head-mounted unit **28** to sense movement of the user's head.

(23) In some embodiments, the tracking sensor **34** includes a light source **42**. In some embodiments, the light source **42** is mounted on the head-mounted unit **28**. In some embodiments, the light source **42** irradiates the field of view of the image-capturing device **36** such that light reflects from the patient marker **38**, the bone marker **60**, and/or the tool marker **40** toward the image-capturing device **36**. In some embodiments, the image-capturing device **36** comprises a monochrome camera with a filter that passes only light in the wavelength band of light source **42**. For example, the light source **42** may be an infrared light source, and the camera may include a corresponding infrared filter. In some embodiments, the patient marker **38**, the bone marker **60**, and/or the tool marker **40** comprise patterns that enable a processor to compute their respective positions, i.e., their locations and their angular orientations, based on the appearance of the patterns in images captured by the image-capturing device **36**. Suitable designs of these markers and methods for computing their positions and orientations are described in the patents and patent applications incorporated herein and cited above.

(24) In addition to or instead of the tracking sensor **34**, the head-mounted unit **28** can include a depth sensor **37**. In the embodiment shown in FIG. 2A, the depth sensor **37** comprises a light source **46** and a camera **43**. In some embodiments, the light source **46** projects a pattern of structured light onto the region of interest (ROI) that is viewed through the one or more displays **30** by a user, such as professional **26**, who is wearing the head-mounted unit **28**. The camera **43** can capture an image of the pattern on the ROI and output the resulting depth data to the processor **32** and/or processor **45**. The depth data may comprise, for example, either raw image data or disparity values indicating the distortion of the pattern due to the varying depth of the ROI. In some embodiments, the processor **32** computes a depth map of the ROI based on the depth data generated by the camera **43**.

(25) In some embodiments, the camera **43** also captures and outputs image data with respect to the markers in system **10**, such as patient marker **38**, bone marker **60**, and/or tool marker **40**. In this case, the camera **43** may also serve as a part of tracking sensor **34**, and a separate image-capturing device **36** may not be needed. For example, the processor **32** may identify patient marker **38**, bone marker **60**, and/or tool marker **40** in the images captured by camera **43**. The processor **32** may also find the 3D coordinates of the markers in the depth map of the ROI. Based on these 3D coordinates, the processor **32** is able to calculate the relative positions of the markers, for example in finding the position of the tool **22** relative to the body of the patient **20**, and can use this information in generating and updating the images presented on head-mounted unit **28**.

(26) In some embodiments, the depth sensor **37** may apply other depth mapping technologies in generating the depth data. For example, the light source **46** may output pulsed or time-modulated light, and the camera **43** may be modified or replaced by a time-sensitive detector or detector array to measure the time of flight of the light to and from points in the ROI. As another option, the light source **46** may be replaced by another camera, and the processor **32** may compare the resulting

images to those captured by the camera **43** in order to perform stereoscopic depth mapping. These and all other suitable alternative depth mapping technologies are considered to be within the scope of the present disclosure.

(27) In the pictured embodiment, system **10** also includes a tomographic imaging device, such as an intraoperative computerized tomography (CT) scanner **41**. Alternatively or additionally, processing system **50** may access or otherwise receive tomographic data from other sources; and the CT scanner itself is not an essential part of the present system. In some embodiments, regardless of the source of the tomographic data, the processor **32** can compute a transformation over the ROI so as to register the tomographic images with the depth maps that it computes on the basis of the depth data provided by depth sensor **37**. The processor **32** can then apply this transformation in presenting a part of the tomographic image on the one or more displays **30** in registration with the ROI viewed through the one or more displays **30**. This functionality is described further hereinbelow with reference to FIG. **4A**.

(28) In some embodiments, in order to generate and present an augmented reality image on the one or more displays **30**, the processor **32** computes the location and orientation of the head-mounted unit **28** with respect to a portion of the body of patient **20**, such as the patient's back. In some embodiments, the processor **32** also computes the location and orientation of the tool **22** with respect to the patient's body. In some embodiments, the processor **45**, which can be integrated within the head-mounted unit **28**, may perform these functions. Alternatively or additionally, the processor **32**, which is disposed externally to the head-mounted unit **28** and can be in wireless communication with the head-mounted unit **28**, may be used to perform these functions. The processor **32** can be part of the processing system **50**, which can include an output device **52**, for example a display, such as a monitor, for outputting information to an operator of the system, and/or an input device **54**, such as a pointing device, a keyboard, or a mouse, to allow the operator to input data into the system.

(29) Alternatively or additionally, users of the system **10** may input instructions to the processing system **50** using a gesture-based interface. For this purpose, for example, the depth sensor **37** may sense movements of a hand **39** of the healthcare professional **26**. Different movements of the professional's hand and fingers may be used to invoke specific functions of the one or more displays **30** and of the system **10**.

(30) In general, in the context of the present description, when a computer processor is described as performing certain steps, these steps may be performed by external computer processor **32** and/or computer processor **45** that is integrated within the head-mounted unit. The processor or processors carry out the described functionality under the control of suitable software, which may be downloaded to system **10** in electronic form, for example over a network, and/or stored on tangible, non-transitory computer-readable media, such as electronic, magnetic, or optical memory.

(31) FIG. **2B** is a schematic pictorial illustration showing details of a head-mounted AR display (HMD) unit **70**, according to another embodiment of the disclosure. HMD unit **70** may be worn by the healthcare professional **26**, and may be used in place of the head-mounted unit **28** (FIG. **1**). HMD unit **70** comprises an optics housing **74** which incorporates a camera **78**, and in the specific embodiment shown, an infrared camera. In some embodiments, the housing **74** comprises an infrared-transparent window **75**, and within the housing, i.e., behind the window, are mounted one or more, for example two, infrared projectors **76**. One of the infrared projectors and the camera may be used, for example, in implementing a pattern-based depth sensor.

(32) In some embodiments, mounted on housing **74** are a pair of augmented reality displays **72**, which allow professional **26** to view entities, such as part or all of patient **20**, through the displays, and which are also configured to present to surgeon images or any other information. In some embodiments, the displays **72** present planning and guidance information, as described above.

(33) In some embodiments, the HMD unit **70** includes a processor **84**, mounted in a processor housing **86**, which operates elements of the HMD unit. In some embodiments, an antenna **88**, may

be used for communication, for example with processor **32** (FIG. **1**).

(34) In some embodiments, a flashlight **82** may be mounted on the front of HMD unit **70**. In some embodiments, the flashlight may project visible light onto objects so that professional is able to clearly see the objects through displays **72**. In some embodiments, elements of the HMD unit **70** are powered by a battery (not shown in the figure), which supplies power to the elements via a battery cable input **90**.

(35) In some embodiments, the HMD unit **70** is held in place on the head of professional **26** by a head strap **80**, and the professional may adjust the head strap by an adjustment knob **92**.

(36) In some embodiments, the HMD may comprise a visor, which includes an AR display positioned in front of each eye of the professional and controlled by the optical engine to project AR images into the pupil of the eye.

(37) In some embodiments, the HMD may comprise a light source for tracking applications, comprising, for example, a pair of infrared (IR) LED projectors, configured to direct IR beams toward the body of patient **20**. In some embodiments, the light source may comprise any other suitable type of one or more light sources, configured to direct any suitable wavelength or band of wavelengths of light. The HMD may also comprise one or more cameras, for example, a red/green/blue (RGB) camera having an IR-pass filter, or a monochrome camera configured to operate in the IR wavelengths. The one or more cameras may be configured to capture images including the markers in system **10** (FIG. **1**).

(38) In some embodiments, the HMD may also comprise one or more additional cameras, e.g., a pair of RGB cameras. In some embodiments, each RGB camera may be configured to produce high-resolution RGB (HR RGB) images of the patient's body, which can be presented on the AR displays. Because the RGB cameras are positioned at a known distance from one another, the processor can combine the images to produce a stereoscopic 3D image of the site being operated on.

(39) In some embodiments, the HMD light source **46** (FIG. **2A**) may comprises a structured light projector (SLP) which projects a pattern onto an area of the body of patient **20** on which professional **26** is operating. In some embodiments, light source **46** comprises a laser dot pattern projector, which is configured to apply to the area structured light comprising a large number (typically between hundreds and hundreds of thousands) of dots arranged in a suitable pattern. This pattern serves as an artificial texture for identifying positions on large anatomical structures lacking fine details of their own, such as the skin and surfaces of the vertebrae. In some embodiments, one or more cameras **43** capture images of the pattern, and a processor, such as processor **32** (FIG. **1**), processes the images in order to produce a depth map of the area. In some embodiments, the depth map is calculated based on the local disparity of the images of the pattern relative to an undistorted reference pattern, together with the known offset between the light source and the camera.

(40) In some embodiments, the projected pattern comprises a pseudorandom pattern of dots. In this case, clusters of dots can be uniquely identified and used for disparity measurements. In the present example, the disparity measurements may be used for calculating depth and for enhancing the precision of the 3D imaging of the area of the patient's body. In some embodiments, the wavelength of the pattern may be in the visible or the infrared range.

(41) In some embodiments, the system **10** (FIG. **1**) may comprise a structured light projector (not shown) mounted on a wall or on an arm of the operating room. In such embodiments, a calibration process between the structured light projector and one or more cameras on the head-mounted unit or elsewhere in the operating room may be performed to obtain the 3D map.

(42) FIG. **3A** is a schematic pictorial illustration showing details of the tool **22**, in accordance with an embodiment of the disclosure. In some embodiments, the tool **22** comprises a handle **162** and a shaft **164**. In some embodiments, the marker **40**, containing a predefined pattern, is disposed on the handle **162** in a fixed spatial relation to shaft **164**. Alternatively, marker **40** may protrude outward from tool **22** to ensure that it is visible to tracking sensor **34** and/or camera **43** (FIG. **2A**).

Additionally or alternatively, the depth sensor **37** generates depth data with respect to tool **22**. Uses of the depth data in tracking and displaying images of tool **22** are described further hereinbelow with reference to FIG. 5.

(43) In some embodiments, based on the image information provided by tracking sensor **34** and/or camera **43**, processor **32** (or processor **45**) is able to compute the position (location and orientation) of marker **40** and to track changes in the position of the marker during the surgical procedure. In some embodiments, using the computed position and the known spatial relation of marker **40** to shaft **164**, processor **32** is thus able to find the orientation angle of a longitudinal axis **166** of shaft **164** and the location of a distal end **168** of the shaft, even when the shaft is inside the patient's body. On this basis, processor **32** generates an image of tool **22**, including shaft **164**, on displays **30** in registration with the ROI viewed through the displays.

(44) In the example shown in FIG. 3A, tool **22** is used in inserting a surgical implant **170** into the body of patient **20**. For the purpose of insertion, implant **170** is mounted on distal end **168** of shaft **164**. In one embodiment, the depth sensor **37** captures depth data with respect to both tool **22** and implant **170**. In some embodiments, the processor **32** uses the depth data in generating and displaying images of both the tool and the implant, as described further hereinbelow with reference to FIG. 6.

(45) FIG. 3B is a schematic pictorial illustration showing details of tool **172**, in accordance with an alternative embodiment of the disclosure. Tool **172** is similar to tool **22** and includes similar components, which are labeled with the same indicator numbers. In contrast to tool **22**, however, tool **172** has a curved shaft **174**. The curvature of shaft **174** may be fixed, or it may be malleable or otherwise bendable, so that the shape of the shaft may vary from one operation to another or even during a single operation.

(46) To identify the actual shape of shaft **174**, processor **32** receives and analyzes a depth map that includes tool **172**. The depth map may be captured using any suitable mapping and/or imaging technique, such as the various techniques described above. In some embodiments, the processor **32** segments the depth map to identify the 3D shape of shaft **174** and thus reconstructs a 3D model of tool **172**. Based on this 3D model, together with the location of tool marker **40**, in some embodiments, the processor **32** is able to compute the location of distal end **168** of shaft **174** even when the distal end is hidden from sight inside the patient's body. In some embodiments, the processor **32** can use the depth data in generating and displaying images of both tool **172** and implant **170**, which professional **26** inserts using tool **172**.

Methods for Image-Guided Surgery Using Depth Sensing

(47) FIG. 4A is a flow chart that schematically illustrates a method for image-guided surgery, in accordance with an embodiment of the disclosure. This method, as well as the methods presented in the figures that follow, is described here with reference to the elements of system **10** (as shown in FIGS. 1, 2A and 3A) for the sake of convenience and clarity. Alternatively, the principles of these methods may be applied, mutatis mutandis, in other sorts of augmented reality systems with suitable head-mounted units and depth sensing capabilities, including specifically the head-mounted units shown in FIG. 2B, as well as in the system shown in FIG. 8.

(48) In some embodiments, the processing system **50** receives a 3D tomographic image, such as a CT image, of at least an ROI within the body, at an image input step **180**. This image may be based on a scan or scans performed before and/or in the course of the surgical procedure. In some embodiments during the procedure, depth sensor **37** generates depth data with respect to the ROI, and processor **32** applies the depth data in computing a depth map of the ROI, at a depth mapping step **182**. For example, the depth map may include the spine of patient **20**, which may be exposed in a surgical procedure.

(49) In some embodiments, the processor **32** registers the 3D volume of the tomographic image from the CT scan with the depth map, at a registration step **184**. For this purpose, for example, processor **32** extracts 3D point clouds from both the tomographic image and the depth map and

then computes a transformation to register one with the other. For instance, during surgery on the spine, the processor may compute the transformation by registering the spine in the depth map with the spine appearing in the tomographic image. Various methods for volume analysis may be used to perform the registration. In one embodiment, processor **32** computes triangular meshes based on the tomographic and depth data and then matches the features of the two meshes.

(50) In some embodiments, the processor **32** applies the transformation computed at step **184** in presenting a part of the tomographic image on displays **30** of head-mounted unit **28** in registration with the ROI viewed through the displays, at a display step **186**. In accordance with several embodiments, to display a tracked tool together with the display of the ROI in step **186**, the depth sensor **37** (e.g., using cameras of the depth sensor **37**) must be calibrated with the tracking system (e.g., tracking sensor **34**). In some embodiments, the processor **32** may use the position of patient marker **38**, as detected by camera **43** and/or tracking sensor **34**, in improving the registration. For example, by identifying the position of patient marker **38** in the depth map, processor **32** may recognize where professional **26** is standing relative to the body of patient **20**. In some embodiments, the processor will then rotate the AR image that is presented on head-mounted unit **28** automatically according to the position of the head-mounted unit relative to the patient marker.

(51) In some embodiments, the processor **32** continues to receive and process depth data output by depth sensor **37** during the surgical procedure. In some embodiments, the processor identifies changes in anatomical structures in the body of patient **20** during the procedure, based on the corresponding changes in the depth map. For example, processor **32** can detect that a part of a bone has been cut or that an implant has been inserted or removed. On this basis, processor **32** can modify the AR image presented on the display of head-mounted unit **28** to reflect the identified change.

(52) FIG. **4B** is a flow chart that schematically illustrates a method for image-guided surgery, in accordance with another embodiment of the disclosure. This embodiment builds on the method of FIG. **4A** and applies the method in detecting movement of the bone marker **60**, and/or movement of a bone anchoring implement or device such as the bone clamp **58** relative to the bone to which it is attached, such as the patient's spine. Identification and measurement of such movement is important since it can affect the overall accuracy of system **10**.

(53) In some embodiments, the processor **32** identifies the location of bone marker **60** and/or the bone anchoring device in an initial depth map that is generated using depth sensor **37**, at a bone marker and/or bone anchoring device identification step **188**. In some embodiments, the depth map may be registered with the tomographic image of the patient's spine using the procedure of step **184** of FIG. **4A**, as described above. In some embodiments, the processor **32** can thus use the depth data in accurately calculating the position of bone marker **60** and/or bone clamp **58** relative the patient's spine.

(54) Subsequently, in the course of the surgery in some embodiments, the depth sensor **37** continues to capture depth data, and processor periodically updates the depth map, at a depth update step **190**. In one embodiment, processor identifies bone marker **60** and/or bone clamp **58** in the updated depth map and checks whether the location of the clamp has shifted relative to the location of spine, as derived from the tomographic image, at a movement detection step **192**. When the location of the clamp in the updated depth map has changed, processor **32** takes a corrective action. For example, the processor **32** may modify the image presented on display **30** to reflect this change, at an image update step **194**. Additionally or alternatively, the processor **32** may issue a warning to the healthcare professional **26** that one of the markers has moved, such as the bone marker **60** or the patient marker **38**, so that the professional can take corrective action if needed.

(55) FIG. **5** is a flow chart that schematically illustrates a method for image-guided surgery, in accordance with yet another embodiment of the disclosure. In some embodiments, the method uses both the 3D shape of the tool **22**, which processor **32** derives from the depth data provided by depth sensor **37**, and the position of the tool marker **40**. The position of the tool marker **40** may be

measured by tracking sensor **34**, or it may alternatively be extracted from images captured by camera **43** of the depth sensor **37**. In either case, in some embodiments, with all the elements used in finding the position of marker **40** and generating depth data with respect to tool **22** being fixed to the head-mounted unit **28** in a known spatial relation, the position of the marker will be consistently registered with the depth map.

(56) In some embodiments, the method of FIG. 5 includes two calibration phases: The first phase includes performing initial calibration prior to the procedure. The second phase provides continuous calibration to update the initial calibration during the procedure. There may be two manners in which the initial calibration may be performed and then updated. In both manners, tool **22** and marker **40** (while the marker is mounted on the tool) are modeled by using depth sensing, for example to identify the trajectory and tip of the tool and the position of the marker with respect to the tool. In some embodiments, the processing system **50** then identifies the tool and marker and determines their relative locations. This function can be carried out using computer vision, for example using function of detection and segmentation, and optionally by applying machine learning techniques, including deep learning. Alternatively or additionally, tracking sensor **34** is used to identify the marker in the 3D model, in which case computer vision may be used only to identify the tool. In the continuous calibration update phase, either or both methods may be used, as well. Although the method shown in FIG. 5 and described below includes both calibration phases, aspects of the method may be applied only in the initial calibration without updating. Furthermore, although some steps in FIG. 5 and in the description below make use of tool marker **40**, the method may alternatively be carried out, *mutatis mutandis*, using only depth sensing without reliance on a tool marker.

(57) In some embodiments, the processor **32** identifies tool marker **40** in an image of tool **22**, at a marker identification step **200**. As noted earlier, the image may be provided either by tracking sensor **34** or by camera **43**. In some embodiments, the processor **32** analyzes the image to derive the position of the tool marker.

(58) In some embodiments, the processor **32** also processes depth data provided by depth sensor **37** in order to generate a depth map of tool **22**, possibly along with other elements in the ROI, at a depth mapping step **202**. In some embodiments, the processor **32** analyzes the shape of tool **22** in the depth map, for example using techniques of 3D image segmentation and/or machine learning that are known in the art. On this basis, for example, the processor **32** may identify shaft **164** (FIG. 3A) and is thus able to find the orientation of axis **166**, corresponding to the “trajectory” of tool **22**, and the location of distal end or tip **168**. When the shaft of the tool is not straight, for example as in tool **172** (FIG. 3B), processor **32** may use the information provided by the depth map in modeling the tool and finding the location and orientation of the distal tip.

(59) Based on the marker position found at step **200** and the depth map analysis of step **202**, in some embodiments, the processor **32** computes a spatial transformation between the position of tool marker **40** and the location and orientation of shaft **164**, at a transformation step **204**. This transformation can be computed on the basis of the position of the marker and the depth data, without requiring prior knowledge of the shape of tool **22** or of the precise position of marker **40** relative to the other parts of the tool. Alternatively, a priori information regarding the position of the marker on the tool may be used to improve the accuracy of the transformation.

(60) Using the tool marker **40** and/or updated depth data, the processor **32** tracks or may receive tracking information about the position of the tool **22** as professional **26** manipulates the tool with the shaft **164** inserted into the body (and thus not visible to depth sensor **37**), at a tracking step **206**. Using the tracked position and the spatial transformation computed at step **204**, processor **32** generates an image of tool **22** on display **30**, in registration with the ROI viewed through the display. The image specifically shows shaft **164**, and particularly distal end **168**, which when placed inside the patient's body may not be directly visible to professional **26**.

(61) As noted earlier, although tool marker **40** is shown in FIG. 3 as a part of handle **162**, in other

embodiments the tool marker may be fixed externally to the tool. In this case, the tool marker is liable to shift relative to the tool. Processor **32** analyzes the depth data provided by depth sensor **37** in order to calibrate the position of the tool marker relative to the tool. Thus, according to some embodiments, the processor may optionally update this analysis periodically at step **206** during the surgical procedure and checks the position to determine whether the tool marker has shifted relative to the tool, at a shift detection step **207**. If so, processor **32** recalibrates the tool position to compensate for the shift, and based on the new calibration, updates the transformation that was computed previously (at step **204**), at a transformation update step **208**. The image presented on display **30** is modified accordingly.

(62) It may also occur that the shape of tool **22** or tool **172** changes in the course of the surgical procedure. For example, the bending angle of shaft **174** may change, or the shaft may even break. Processor **32** analyzes the depth data provided by depth sensor **37** in order to determine whether the shape of the tool has changed, at a shape change detection step **209**. If so, processor **32** recalibrates the tool shape to compensate for the change, and based on the new calibration, updates the transformation that was computed previously (at step **204**), at transformation update step **208**. The image of the tool on display **30** is updated, as well.

(63) FIG. **6** is a flow chart that schematically illustrates a method for image-guided surgery, in accordance with a further embodiment of the disclosure. This embodiment extends the method of FIG. **5** to deal with the shape and position of implant **170**, in addition to tool **22**. Although FIG. **3** shows implant **170** as a screw (for example a pedicle screw), the present method is equally applicable to implants of other types, such as stents, cages, and interbody devices. The present method is advantageous in that it enables processor **32** to extract the shapes of implants without requiring prior knowledge of the implant characteristics, and to use the extracted shapes in presenting virtual images of the implants on display **30**.

(64) As an initial step, before inserting implant **170** into the body of patient **20**, processor **32** generates (or accesses if depth maps were previously generated) depth maps of both implant **170** and tool **22**, at a depth mapping step **210**. For this purpose, for example, a user of system **10** may be prompted to operate depth sensor **37** to capture depth data with respect to implant **170** by itself and also with respect to tool **22** with implant **170** mounted on distal end **168**. Processor **32** analyzes the depth data to identify the shape of implant **170**. Based on this shape and on the depth map of implant **170** mounted on tool **22**, processor **32** computes a spatial transformation between the position of the tip of tool **22** and the location and orientation of implant **170**, at a registration step **212**. A spatial transformation between the position of marker **40** and the location and orientation of implant **170** may be computed therefrom. Alternatively or additionally, a transformation between marker **40** and the tip of tool **22** may have been computed at an earlier phase, for example at a calibration phase as described above.

(65) Following these steps, professional **26** uses tool **22** to insert implant **170** into the body of patient **20**. Processor **32** tracks the position of marker **40** as professional **26** manipulates shaft **164** of tool **22** within the body. According to some aspects, such tracking information is accessed by processor **32**. Using the tracked position, the spatial transformation computed at step **212**, and the shape of the implant that was identified at step **210**, processor **32** generates a virtual image on display **30** showing the implant within the body and/or its navigation, in registration with the ROI viewed through the display, at an image generation step **214**.

(66) After implant **170** has been inserted in place, processor **32** may continue processing the depth data provided by depth sensor **37** in order to identify and locate the implant, at an implant tracking step **216**. For example, processor **32** may identify the head of a screw in the depth map as a cue to the location and orientation of the screw. The display on head-mounted unit **28** can be modified to show the implant in the proper location. The information regarding the location and orientation of the implant can also be useful in planning subsequent stages of the procedure, such as insertion of rods between the screws.

(67) FIG. 7 is a flow chart that schematically illustrates a method for image-guided surgery, in accordance with an additional embodiment of the disclosure. This method is applicable particular in osteotomies, to assist in visualizing the portion of a bone that is removed in the procedure. In such a procedure, processing system **50** accesses a plan made by the surgeon to remove a certain volume of the bone in question, for example a part of one or more vertebrae, at a planning step **220**. The present method can assist the surgeon in comparing the volume of the bone that has been cut out to the planned volume.

(68) For this purpose, prior to cutting of the bone, the processor **32** processes depth data generated by depth sensor **37** so as to identify the 3D shape of the bone, for example by generating a point cloud. The 3D cutting plan can be overlaid on the patient's actual anatomy (without displaying patient image data), optionally in a semi- or partially-transparent manner. For example, the top plane of the plan can be overlaid on the patient in alignment with and oriented according to the patient anatomy. The plan may be used as a guide for cutting.

(69) As the surgeon cuts the bone, the outline or indication of a plane of the bone to be removed according to plan changes according to the cutting tool tip location, for example based on depth of penetration. The plan outline or plane can be displayed from a point of view defined by the tool orientation, as determined by tracking the tool tracking. This mode is especially compatible with a "Tip View" mode, in which the patient spine model displayed on the near-eye display, generated based on CT data, changes according to the tool tip location and such that the upper surface of the model is the upper plane defined by the tool orientation and tip location. For example, the model may be cut up to a plane orthogonal to the tool trajectory or longitudinal orientation such that only a portion of the model is displayed.

(70) It is also possible to display or indicate what was done already or the portion of the procedure already performed (e.g., what portion of the bone was already cut and what portion of the bone is left to be cut). The already-cut portion may be indicated on the plan, for example in a different color, or augmented on the image or on the actual patient anatomy or the plan may be updated to show only the remaining bone, and may also show when a portion of the bone was not cut according to plan. Tracking of the cutting may be performed based on tool tip tracking or by depth sensing.

(71) To track cutting using depth sensing, a first depth image of the bone is captured prior to cutting. During the cutting, additional depth images are captured. The capturing may be performed on user request or automatically, continuously or at predefined time interval. Each depth image is compared to the previous one, and processor **32** identifies whether a bone portion was removed or not, i.e., whether cutting was performed. When a difference in the bone volume is identified, the portion of bone that was removed may be indicated on or compared to the plan and displayed to the user. The depth sensor and the tracking system may be calibrated for this purpose. Alternatively, or additionally, the depth maps may be registered with the CT model, for example using feature matching. The calibration and registration process may allow comparison between the different depth maps, the CT model, and the plan.

(72) After the bone has been cut, in some embodiments, the processor **32** accesses and processes new depth data in order to identify the modified 3D shape of the bone. Based on the difference between the 3D shapes, processor **32** identifies the portion of the bone that was removed, at an excision identification step **222**. In some embodiments, the processor **32** can then display an image showing the part of the bone that was removed in the surgical procedure, at a display step **224**. The surgeon can compare this image to the plan in order to verify that the osteotomy was completed according to plan. In some embodiments, the processor **32** can display both images, i.e., of the removed bone volume and of the planned volume, simultaneously to facilitate comparison between the two. The images may be displayed adjacent to or overlaid on one another.

(73) Alternatively, processor **32** may display to the surgeon only the removed portion of the bone, without comparison to a plan. The processor may thus demonstrate the removed volume and assist

in confirming the procedure or in deciding whether a correction or a further operation is required, for example. Additionally or alternatively, in cases in which an implant is to be placed in the body in place of the removed portion of the bone, the surgeon and/or processor **32** may use the model of the removed bone portion to select a suitable implant or to determine whether a particular implant is suitable. On this basis, a suitable implant may be selected from a database, for example. When comparing the removed bone volume to a specific implant, size data may be provided with respect to the implant, or it may be generated using the depth sensing techniques described above.

(74) In some embodiments, the processor **32** generates a model of the removed portion of the bone, which can be used as an input for an implant printing device (such as a 3D printer). The implant may thus be designed based on the model of the removed portion of bone.

(75) In one embodiment, the image and/or model of the ROI that is presented on the AR display, such as the patient spine model, dynamically changes according to the cutting performed and based on the tracking of cutting that is described above. Thus, if a drill is used, for example, then holes may be formed in the model correspondingly, based on tool tracking, for example. The model may be updated in this manner during the procedure and during cutting. At the end of the procedure, the user may be presented with a model showing the entire bone portion removed.

Alternative Embodiment

(76) FIG. **8** is a schematic pictorial illustration of a system **240** for image-guided surgery, in accordance with an alternative embodiment of the invention. System **240** is similar in design and operation to system **10** (FIG. **1**), except that system **240** does not use the sorts of patient marker and clamp marker that were described above, with the accompanying tracking system. Rather, system **240** comprises an array of fiducial markers **242**, which are fixed to the back of patient **20**, for example using a suitable adhesive. Fiducial markers **242** comprises 3D elements, such as metal beads, which are both radiopaque and visible to cameras, such as camera **43** (FIG. **2A**).

(77) Prior to the surgical procedure, markers **242** are fixed to the back of patient **20**, and a CT scan of the patient is performed, for example using CT scanner **41** (FIG. **1**). Markers **242** will appear in the resulting CT images along with the patient's skeleton. During the surgery, depth sensor **37** on head-mounted unit **28** captures depth maps of the patient's back, including markers **242**. Processor **32** analyzes the depth maps to find the 3D coordinates of markers **242**. The processor then matches the locations of markers **242** in the CT image data with the 3D coordinates of the markers in the depth map, and thus derives the appropriate transformation to register the CT image data with the actual patient anatomy viewed by professional **26**. Based on this registration, processor **32** presents parts of the CT images on displays **30**, overlaid on the patient anatomy.

(78) The other features and applications of depth mapping in image-guided surgery that are described above may likewise be applied, mutatis mutandis, in system **240**.

(79) Although the drawings and embodiments described above relate specifically to surgery on the spine, the principles of the present disclosure may similarly be applied in other sorts of surgical procedures, such as operations performed on the cranium and various joints, as well as dental surgery. It will thus be appreciated that the embodiments described above are cited by way of example, and that the present disclosure is not limited to what has been particularly shown and described hereinabove. Rather, the scope of the present disclosure includes both combinations and subcombinations of the various features described hereinabove, as well as variations and modifications thereof which would occur to persons skilled in the art upon reading the foregoing description and which are not disclosed in the prior art.

(80) Indeed, although the systems and processes have been disclosed in the context of certain embodiments and examples, it will be understood by those skilled in the art that the various embodiments of the systems and processes extend beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the systems and processes and obvious modifications and equivalents thereof. In addition, while several variations of the embodiments of the systems and processes have been shown and described in detail, other modifications, which are

within the scope of this disclosure, will be readily apparent to those of skill in the art based upon this disclosure. It is also contemplated that various combinations or sub-combinations of the specific features and aspects of the embodiments may be made and still fall within the scope of the disclosure. It should be understood that various features and aspects of the disclosed embodiments can be combined with, or substituted for, one another in order to form varying modes of the embodiments of the disclosed systems and processes. Any methods disclosed herein need not be performed in the order recited. Thus, it is intended that the scope of the systems and processes herein disclosed should not be limited by the particular embodiments described above.

(81) It will be appreciated that the systems and methods of the disclosure each have several innovative aspects, no single one of which is solely responsible or required for the desirable attributes disclosed herein. The various features and processes described above may be used independently of one another or may be combined in various ways. All possible combinations and sub-combinations are intended to fall within the scope of this disclosure.

(82) Certain features that are described in this specification in the context of separate embodiments also may be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment also may be implemented in multiple embodiments separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination may in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination. No single feature or group of features is necessary or indispensable to each and every embodiment.

(83) Each of the processes, methods, and algorithms described in the preceding sections may be embodied in, and fully or partially automated by, code modules executed by one or more computer systems or computer processors comprising computer hardware. The code modules may be stored on any type of non-transitory computer-readable medium or computer storage device, such as hard drives, solid state memory, optical disc, and/or the like. The systems and modules may also be transmitted as generated data signals (for example, as part of a carrier wave or other analog or digital propagated signal) on a variety of computer-readable transmission mediums, including wireless-based and wired/cable-based mediums, and may take a variety of forms (for example, as part of a single or multiplexed analog signal, or as multiple discrete digital packets or frames). The processes and algorithms may be implemented partially or wholly in application-specific circuitry. The results of the disclosed processes and process steps may be stored, persistently or otherwise, in any type of non-transitory computer storage such as, for example, volatile or non-volatile storage.

(84) The various features and processes described above may be used independently of one another or may be combined in various ways. All possible combinations and sub-combinations are intended to fall within the scope of this disclosure. In addition, certain method or process blocks may be omitted in some implementations. The methods and processes described herein are also not limited to any particular sequence, and the blocks or states relating thereto can be performed in other sequences that are appropriate. For example, described blocks or states may be performed in an order other than that specifically disclosed, or multiple blocks or states may be combined in a single block or state. The example blocks or states may be performed in serial, in parallel, or in some other manner. Blocks or states may be added to or removed from the disclosed example embodiments. The example systems and components described herein may be configured differently than described. For example, elements may be added to, removed from, or rearranged compared to the disclosed example embodiments.

(85) Conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more embodiments or

that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment.

(86) The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. In addition, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. In addition, the articles “a,” “an,” and “the” as used in this application and the appended claims are to be construed to mean “one or more” or “at least one” unless specified otherwise. Similarly, while operations may be depicted in the drawings in a particular order, it is to be recognized that such operations need not be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. Further, the drawings may schematically depict one or more example processes in the form of a flowchart. However, other operations that are not depicted may be incorporated in the example methods and processes that are schematically illustrated. For example, one or more additional operations may be performed before, after, simultaneously, or between any of the illustrated operations. Additionally, the operations may be rearranged or reordered in other embodiments. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments, and it should be understood that the described program components and systems may generally be integrated together in a single software product or packaged into multiple software products. Additionally, other embodiments are within the scope of the following claims. In some cases, the actions recited in the claims may be performed in a different order and still achieve desirable results.

(87) As used herein “generate” or “generating” may include specific algorithms for creating information based on or using other input information. Generating may include retrieving the input information such as from memory or as provided input parameters to the hardware performing the generating. Once obtained, the generating may include combining the input information. The combination may be performed through specific circuitry configured to provide an output indicating the result of the generating. The combination may be dynamically performed such as through dynamic selection of execution paths based on, for example, the input information, device operational characteristics (for example, hardware resources available, power level, power source, memory levels, network connectivity, bandwidth, and the like). Generating may also include storing the generated information in a memory location. The memory location may be identified as part of the request message that initiates the generating. In some implementations, the generating may return location information identifying where the generated information can be accessed. The location information may include a memory location, network locate, file system location, or the like.

(88) Any process descriptions, elements, or blocks in the flow diagrams described herein and/or depicted in the attached figures should be understood as potentially representing modules, segments, or portions of code which include one or more executable instructions for implementing specific logical functions or steps in the process. Alternate implementations are included within the scope of the embodiments described herein in which elements or functions may be deleted, executed out of order from that shown or discussed, including substantially concurrently or in reverse order, depending on the functionality involved, as would be understood by those skilled in the art.

(89) All of the methods and processes described above may be embodied in, and partially or fully automated via, software code modules executed by one or more general purpose computers. For example, the methods described herein may be performed by the processors **32**, **45** described herein and/or any other suitable computing device. The methods may be executed on the computing

devices in response to execution of software instructions or other executable code read from a tangible computer readable medium. A tangible computer readable medium is a data storage device that can store data that is readable by a computer system. Examples of computer readable mediums include read-only memory, random-access memory, other volatile or non-volatile memory devices, CD-ROMs, magnetic tape, flash drives, and optical data storage devices.

(90) Documents incorporated by reference in the present patent application are to be considered an integral part of the application except that, to the extent that any terms are defined in these incorporated documents in a manner that conflicts with definitions made explicitly or implicitly in the present specification, only the definitions in the present specification should be considered.

(91) It should be emphasized that many variations and modifications may be made to the above-described embodiments, the elements of which are to be understood as being among other acceptable examples. All such modifications and variations are intended to be included herein within the scope of this disclosure. The foregoing description details certain embodiments. It will be appreciated, however, that no matter how detailed the foregoing appears in text, the systems and methods can be practiced in many ways. As it is also stated above, it should be noted that the use of particular terminology when describing certain features or aspects of the systems and methods should not be taken to imply that the terminology is being re-defined herein to be restricted to including any specific characteristics of the features or aspects of the systems and methods with which that terminology is associated. While the embodiments provide various features, examples, screen displays, user interface features, and analyses, it is recognized that other embodiments may be used.

Claims

1. A head-mounted system for image-guided surgery, comprising: a head-mounted unit, comprising a see-through augmented-reality display and a depth sensor, which is configured to generate depth data with respect to a body of a patient that is viewed through the display by a user wearing the head-mounted unit, the depth data including data for both a region of interest (ROI) within the body of the patient and an array of radiopaque fiducial markers fixed externally to the body of the patient; and a processor, which is configured to: receive a three-dimensional (3D) tomographic image of the body of the patient; compute a depth map based on the depth data generated by the depth sensor, the depth map including both the ROI within the body of the patient and the array of radiopaque fiducial markers fixed externally to the body of the patient; compute a transformation over the ROI so as to register the tomographic image with the depth map; and apply the transformation in presenting a part of the tomographic image on the display in registration with the ROI viewed through the display, wherein the ROI within the body of the patient that is included in the depth map includes a spine of the patient, which is exposed in a surgical procedure, wherein the processor is configured to compute the transformation by registering the spine in the depth map with the spine appearing in the tomographic image, wherein the tomographic image comprises a CT scan of the patient which was performed with the array of radiopaque fiducial markers fixed externally to the body of the patient, wherein the processor is configured to identify respective 3D coordinates of the fiducial markers in the depth map and to register the CT scan with the ROI viewed through the display by matching the fiducial markers in the CT to the respective 3D coordinates, and wherein the processor is configured to compute an updated depth map in a course of a surgery to enable detection of a change in the ROI within the body of the patient based on the updated depth map.

2. The system of claim 1, wherein the ROI comprises a bone of the body to which an anchoring device is coupled, and wherein the processor is further configured to: identify a location of the anchoring device in the depth map; detect a change in the location of the anchoring device in the updated depth map; and take a corrective action responsively to the change.

3. The system of claim 2, wherein the corrective action comprises modifying a presentation on the display responsively to the change in the location of the anchoring device.
4. The system of claim 1, wherein the processor is configured to: process the updated depth map so as to identify a change in an anatomical structure in the body of the patient during the surgery; and modify the image presented on the display responsively to the identified change.
5. The system of claim 1, wherein the processor is configured to: process the updated depth map so as to identify an implant inserted into the body of the patient during the surgery; and modify the image presented on the display responsively to the identified implant.
6. The system of claim 1, wherein the 3D tomographic image is a computed tomographic image.
7. The system of claim 1, wherein the depth sensor comprises a pattern projector and a camera configured for structured-light depth mapping.
8. The system of claim 7, wherein the pattern projector comprises a laser dot pattern projector configured to apply structured light comprising dots arranged in a pattern.
9. The system of claim 8, wherein the pattern is a pseudorandom pattern of dots.
10. The system of claim 9, wherein the depth map is calculated based on a local disparity of images of the pattern captured by the camera relative to an undistorted reference pattern, together with a known offset between the pattern projector and the camera.
11. The system of claim 1, wherein the depth sensor comprises a pair of cameras configured for stereoscopic depth mapping.
12. The system of claim 1, wherein the depth sensor comprises a beam projector and one or more detectors configured for time-of-flight measurement.
13. The system of claim 1, wherein the head-mounted unit comprises a head strap.
14. The system of claim 1, wherein the head-mounted unit comprises a visor.
15. The system of claim 1, wherein the processor is further configured to: process the depth map to identify a 3D shape of a vertebrae bone prior to a cutting of the vertebrae bone; process the updated depth map to identify a modified 3D shape of the vertebrae bone after the cutting of the vertebrae bone; based on a difference between the 3D shape and the modified 3D shape, identify a portion of the vertebrae bone that was removed; and present an image on the display showing the identified portion of the vertebrae bone that was removed.
16. A head-mounted system for image-guided surgery comprising: a see-through augmented-reality display disposed so as to be viewable by a user over a body of a patient; a depth sensor configured to generate depth data with respect to the body of the patient, the depth data including data for both a region of interest (ROI) within the body of the patient and one or more markers positioned external to the body of the patient; and a processor and a memory for storing instructions that, when executed by the processor cause the system to: receive a three-dimensional (3D) tomographic image of the body of the patient; determine a depth map based at least in part on the depth data, the depth map including both the ROI within the body of the patient and the one or more markers positioned external to the body of the patient; determine a transformation over the ROI so as to register the 3D tomographic image with the depth map; display at least a part of the 3D tomographic image on the see-through augmented-reality display in registration with the ROI based at least in part on the transformation; and determine an updated depth map in a course of a surgery to enable detection of a change in the ROI within the body of the patient based on the updated depth map.
17. The system of claim 16, wherein: the 3D tomographic image is a computed tomographic image; the depth sensor comprises a pattern projector and a camera configured for structured-light depth mapping; and the pattern projector comprises a laser dot pattern projector configured to apply structured light comprising dots arranged in a pattern.
18. The system of claim 17, wherein the pattern is a pseudorandom pattern of dots.
19. The system of claim 18, wherein the depth map is calculated based on a local disparity of images of the pattern captured by the camera relative to an undistorted reference pattern, together

with a known offset between the pattern projector and the camera.

20. The system of claim 16, further comprising a head-mounted unit comprising the see-through augmented-reality display.
